SDI Investment Property Marketplace

System Documentation

What has been built, how to run it, and who can see what

Release	`v0.9.13`
Repository	`github.com/neilgreene/property-management`
Branch	`main`
Commits	13
Database	PostgreSQL 16 (row-level security; requires 15+)
Application	Node.js 18+ (no framework, no runtime dependencies beyond `pg`)
Automated checks	63 worker tests plus 4 SQL walkthroughs, all passing
Deployment status	Local development only. Not deployed, not internet-facing.

Every fact in this document was extracted from the repository and from a freshly built database, not written from memory. Regenerate it with `python3 docs/generate_system_documentation.py`.

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Contents

1. What This Is

A working system for a private investment property marketplace: vetted properties listed with financial analysis, browsable without revealing addresses, with the full detail unlocked only after an investor signs and pays the platform fee agreement. Agents see only their own assignments. Staff see everything.

1.1 The problem it solves first

The commercial model depends on one thing holding: an unpaid visitor must not be able to obtain a property's street address. Everything else is ordinary software; that is not. So the visibility rules are enforced **inside the database**, by row-level security policies and column grants, rather than by application code.

This matters because it changes what an attacker has to defeat. If the rule lives in application code, anyone who finds an unguarded query, an API bug, or a direct database connection gets the address. Here, the address is withheld by the database itself: a caller who writes their own SQL, bypasses the application entirely, and connects directly still cannot read it.

1.2 Three visibility bands

Band	Contains	Released to	Enforced by
1 — Public	City, price, cap rate, NOI, beds, baths, year built	Anyone, including anonymous visitors	RLS policy per role
2 — Gated	Street address, unit, parcel number, seller disclosure, true coordinates	Investors who signed AND paid; the assigned agent; staff	Masking in a `security_invoker` view, keyed on a data predicate
3 — Internal	Acquisition cost, source channel, staff notes, margin	Staff only	Column-level GRANT (a hard ACL)

Coordinates degrade rather than disappear. An ungated viewer gets a deterministic offset of roughly one kilometre, seeded on the property id, so a map still renders a neighbourhood and repeated page loads cannot be averaged to recover the true point.

2. How To Run It

2.1 Requirements

Two ways to run this, with different prerequisites. The Docker route needs one thing installed; the local route needs three, but gives faster iteration.

Route A — Docker (recommended for a first run)

Requirement	Version	Notes
Docker Engine	20.10 or later	Any host: Linux, macOS, Windows with WSL2, or a Proxmox VM
Docker Compose	v2 (the `docker compose` subcommand)	Not the older standalone `docker-compose` binary; this file uses v2 syntax including profiles

Requirement	Version	Notes
Disk	about 1.5 GB	Chiefly the two base images
Memory	1 GB free	PostgreSQL and two small Node processes
Network at build time	outbound HTTPS	Pulls images from Docker Hub and `pg` from the npm registry. Nothing else is fetched.

What is inside the Compose file

Service	Image	Resolves to	Purpose	Port
db	`postgres:16`	PostgreSQL 16.x — tested on 16.13	Database; loads every `sql/` file on first start	5432
web	`node:22-alpine`	Node 22.x — tested on 22.22.2	The demo interface	3000
worker	`node:22-alpine`	Node 22.x — tested on 22.22.2	GoHighLevel integration. Behind the `worker` profile; not started by default	3001

Both tags float within their major version, so a pull next month may bring a newer patch release. That is the right default for development — it picks up security fixes — but for production pin them to a digest so a rebuild produces the same image that was tested. The versions above are what this system has actually been verified against.

Only one runtime dependency is installed at all: `pg` (resolves to 8.23.x under the declared `^8.13.1`). There is no web framework, no ORM, no build step and no bundler. That is deliberate — it keeps the dependency audit trivial and removes a class of supply-chain exposure — and it is worth preserving.

Route B — local install

Requirement	Version	Why that floor
PostgreSQL	16 or later	`security_invoker` views need 15+; developed and tested on 16. Below 15 a view over an RLS table silently runs as its owner and bypasses the caller's policies, which would defeat the entire model.
Node.js	18 or later	Uses the built-in test runner and global `fetch`, both stable from 18. Tested on 22.
npm	9 or later	Ships with Node 18+
`psql` and `createdb`	matching the server	`run.sh` calls them as your own user, which is the default for Postgres.app and Homebrew

Optional — only for rebuilding this document

Requirement	Version	Used by
Python	3.9 or later — tested on 3.11	Both PDF generators
`reportlab`	4.x or later — tested on 5.0.1	`pip install reportlab`

Requirement	Version	Used by
`pypdfium2`	any	Optional. Only to render pages for visual checking

Nothing in the running system needs Python. It is only used to regenerate the two PDFs in `docs/`.

Network access

When	Needs	For
Build	Outbound HTTPS to Docker Hub and `registry.npmjs.org`	Images and the one npm package
Runtime, core system	None	The database, demo and tests are entirely self-contained and run air-gapped
Runtime, worker	Outbound HTTPS to `services.leadconnectorhq.com`	The GoHighLevel API
Runtime, webhooks	Inbound HTTPS on a public address	Only for receiving GoHighLevel deliveries. Everything else works without it.

The distinction in that last table matters for choosing where to run. The whole system minus webhook receipt works with no inbound access at all, which is why a Proxmox VM with no port forwarding is a perfectly good staging environment.

2.2 Docker — nothing installed but Docker

```
git clone https://github.com/neilgreene/property-management
cd property-management
docker compose up
```

Builds the database from `sql/`, seeds it, and serves the demo at **http://localhost:3000**. PostgreSQL is exposed on `localhost:5432` (database `sdi`, user `postgres`, password `postgres`). `docker compose down -v` discards the database.

2.3 Local PostgreSQL 16+ and Node 18+

```
./run.sh
```

Loads all eleven schema files, runs all four SQL walkthroughs, runs the 63 worker tests, then starts the demo. It assumes `psql` and `createdb` work as your own user, which is the default for Postgres.app and Homebrew.

2.4 Individual pieces

Command	What it does
`psql -d sdi -f sql/05_tests.sql`	Security walkthrough: 11 checks, 5 of them attacks
`psql -d sdi -f sql/07_ghl_tests.sql`	GoHighLevel bridge: 7 checks

Command	What it does
`psql -d sdi -f sql/10_review_tests.sql`	Review queue actions: 7 checks
`psql -d sdi -f sql/14_pipeline_tests.sql`	Deal visibility and stage history: 9 checks
`cd worker && npm test`	63 unit and end-to-end tests

2.5 The integration worker

Deliberately not started by a bare `docker compose up`, because it needs real GoHighLevel credentials and there is no point running it without them.

```
GHL_TOKEN=... GHL_LOCATION_ID=... docker compose --profile worker up
```

Variable	Where it comes from
`GHL_TOKEN`	GoHighLevel: Settings → Private Integrations. Scoped to an entire sub-account.
`GHL_LOCATION_ID`	The sub-account id, visible in the GoHighLevel URL
`GHL_WEBHOOK_PUBLIC_KEY`	GoHighLevel's published webhook verification key (PEM)

The token is read from the environment and is never written into any file in the repository. It grants access to the whole sub-account — contacts, invoices, transactions — so it must never reach browser-side code.

3. Users and Access

3.1 Signing in

Authentication is built. A person signs in with an email and a password, gets a session cookie, and the web tier resolves that cookie to a person id and a database role for the duration of one transaction.

Concern	How it is handled
Password storage	`scrypt` with per-password salt and explicit cost parameters. Node's built-in implementation — no dependency
Failed sign-in	One message and one shape for every failure. Distinguishing “no such account” from “wrong password” is an enumeration oracle, so the failure path does the same hashing work as the success path
Repeated failures	The account locks after five. Setting a new password clears the lock
Sessions	A random token; only its SHA-256 is stored, so the session table is not a set of usable credentials. Changing a password revokes every existing session
Credential tables	`core.credential` and `core.session` are RLS-forced with **no policy at all** — every direct read is refused for every role, including the owner. They are reachable only through `SECURITY DEFINER` functions

Authentication changed nothing beneath it. Not one policy, view or grant was modified to add it: the database contract was always “here is a role and an actor id”, and a session now supplies those instead of a dropdown. That is the payoff of having put the authorisation model in the database first.

The demo persona switcher still exists, but it is off unless `DEMO_PERSONAS=1` is set. A dropdown that hands out an admin session must not be reachable by accident.

Still missing: password reset, email delivery, and multi-factor. A person who forgets a password needs a staff member to set a new one.

3.2 Seeded people

Created by `sql/04_seed.sql`. These are demonstration records, not real people; the addresses and financials attached to them are invented. Every one of them has the password `demo1234`, set by `sql/17_demo_passwords.sql` — a file that belongs in a demo and nowhere else.

Name	Role	Email	Fee agreement	Brand	What they demonstrate
Jessica Pool	admin	`jpool2@yahoo.com`	n/a	BRAND_A	Full staff access, all bands
Dan Beitor	admin	`dan@example.com`	n/a	BRAND_A	Full staff access, all bands
Tom Bradbury	agent	`tom@example.com`	n/a	BRAND_A	4 assigned properties, incl. an unpublished draft
Priya Raman	agent	`priya@example.com`	n/a	BRAND_A	A second agent, to prove agents are isolated

Name	Role	Email	Fee agreement	Brand	What they demonstrate
Ruth Okonkwo	investor	`ruth@example.com`	signed	BRAND_A	Gate open: sees addresses and true coordinates
Marcus Pell	investor	`marcus@example.com`	not signed	BRAND_A	Gate shut: same query, address withheld
Ines Duarte	investor	`ines@example.com`	signed	KAVADOO	The second brand, at concierge pricing

Ruth and Marcus are the pair that carries the argument: identical role, identical SQL, and the only difference between them is one timestamp in a data column. The address appears for one and not the other with no application logic involved at all.

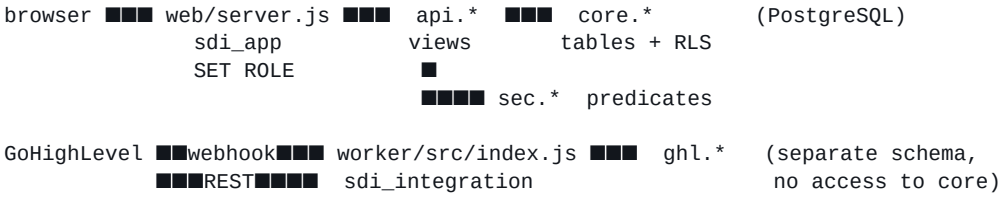
3.3 Database roles

Application roles are created `NOLOGIN` and passwordless on purpose. `sdi_app` is `NOINHERIT`, so it holds no privileges of its own and must assume exactly one persona role per transaction — the application cannot forget to assume a role and accidentally run with more authority than it should.

Role	Purpose	Login	Demo password
`sdi_app`	The only role the web tier connects as	yes	`demo_app_pw`
`sdi_public`	Anonymous visitors. Band 1 only	no	assumed via SET ROLE
`sdi_investor`	Investors. Band 2 if the gate is open	no	assumed via SET ROLE
`sdi_agent`	Agents. Band 2 on assigned properties	no	assumed via SET ROLE
`sdi_admin`	Staff. All three bands	no	assumed via SET ROLE
`sdi_integration`	The GoHighLevel worker. No access to core at all	yes	`demo_int_pw`
`sdi_test_admin`	Test fixtures only. BYPASSRLS	yes	`demo_test_pw`

Those passwords come from `sql/99_local_logins.sql`, exist so the demo stack is connectable, and are published in a public repository. They are worth exactly nothing and that file must not be loaded anywhere that matters. `sdi_test_admin` carries `BYPASSRLS` and belongs only in a test database.

4. Architecture



4.1 Four schemas, and the separation is load-bearing

Schema	Holds	Who has USAGE
`core`	Base tables. Properties, people, deals, assignments	No application role. Name resolution fails before any ACL or policy is consulted, so there is no path around the views.
`sec`	Security predicates, definer-rights	All persona roles. They live here because a `security_invoker` view resolves the functions it calls as the caller too, not just the tables.
`api`	Masking views and the write path	All persona roles. The only surface the application reads.
`ghl`	The GoHighLevel bridge	`sdi_integration` only. The web tier cannot enumerate CRM contacts, invoices or transactions.

Brand is a lens, not a property attribute. Both brands read the same `core.property` rows; `core.property_brand` decides publication and price per brand. Adding the second brand is rows in one table, not a schema refactor.

4.2 Tables

Schema	Tables
`core`	`brand`, `person`, `property`, `property_brand`, `property_assignment`, `saved_property`, `pipeline`, `pipeline_stage`, `deal`, `deal_stage_history`
`ghl`	`id_map`, `webhook_event`, `transaction`, `fee_agreement`, `sync_state`, `outbox`, `review_queue`, `reviewable_field`

All ten `core` tables have row-level security enabled and forced; none of the `ghl` tables do, because that schema is reachable only by the integration worker and by nothing else.

Every column of every table is defined in Appendix A, with its type, nullability, default, key role and foreign key target. That appendix is generated from the live database rather than transcribed, so it cannot describe a schema the system does not actually have.

5. The API

5.1 Database views — what the application reads

View	Returns	Granted to
`api.property`	Properties with band 2 masked or released per caller	public, investor, agent, admin
`api.property_internal`	Band 3: acquisition cost, source, notes, margin	admin only
`api.my_saved`	The caller's own saved properties	investor, admin
`api.deal`	Deals the caller is party to, with stage and age	investor, agent, admin
`api.deal_history`	Stage transitions for those deals	investor, agent, admin
`api.review_open`	Inbound CRM edits awaiting a decision	admin only

Note that `api.deal` is not granted to `sdi_public` at all. A deal is never public, at any stage.

5.2 Callable functions

Function	Does	Caller
`api.save_property(uuid)`	Saves a property to the caller's list, rejecting anything they cannot see	investor
`api.review_decide(bigint, text)`	Accept or reject a queued CRM edit; accepting writes an allowlist of columns only	admin
`api.security_invariants()`	Returns zero rows, or names what is broken	admin
`ghl.apply_fee_agreement(text)`	Opens the gate for a person, if the document is completed AND paid	integration worker

The `sec.*` predicates (`can_see_address`, `can_see_deal`, `is_assigned`, `actor`, `jitter` and others) are internal. They are the shared source of truth that policies, views and write functions all consult, so a rule cannot drift between the place it is read and the place it is enforced.

5.3 HTTP endpoints

Service	Method and path	Purpose
Marketplace	`GET /`	The marketplace: filters, map, cards, drill-down
Marketplace	`POST /api/login` · `/api/logout`	Sign in and out. One response shape for every failure
Marketplace	`GET /api/whoami`	Who this session is, and whether personas are enabled
Marketplace	`GET /api/listings?<filters>`	Filtered listings, plus facet ranges and the vocabularies for the dropdowns
Marketplace	`GET /api/property?id=`	One listing in full, with its photographs. 404 when the row policy hides it
Marketplace	`POST|DELETE /api/favorite`	Add or remove a favourite
Marketplace	`GET /api/favorites`	The caller's favourites, as full listing cards

Service	Method and path	Purpose
Marketplace	`GET|POST|DELETE /api/saved-search`	List, upsert by name, delete
Marketplace	`POST /api/saved-search/run`	Replay a saved search; returns its criteria and records the run
Marketplace	`POST /api/parse`	Plain English to a bounded criteria object — see 7.5
Marketplace	`GET /media/<id>/<kind>.svg`	Generated listing illustration
Marketplace	`GET /api/view · /api/probe`	The teaching endpoints: one persona's whole payload, and a direct base-table read that is refused
Worker	`POST /webhooks/ghl`	Receives a GoHighLevel delivery, verifies its signature
Worker	`GET /healthz`	Queue depth: pending events, bad signatures, outbox backlog, open reviews

The marketplace runs on port 3000, the worker on 3001. The marketplace authenticates; the worker does not, and neither should be exposed to a network until the deployment gaps in section 12 are addressed.

6. The GoHighLevel Integration

Built against GoHighLevel's official OpenAPI specifications. The full API contract is documented separately in `docs/GHL-Interface-Specification.pdf` (17 pages).

6.1 What the worker does

Component	Responsibility
`ghlClient.js`	HTTP client. Sets the required `Version: 2021-07-28` header once, paces under the 100-request/10-second limit, retries 429 and 5xx, never retries 403
`signature.js`	Verifies webhook signatures against GoHighLevel's published RSA public key
`webhookReceiver.js`	Records deliveries, deduplicating on `webhookId` and rejecting stale ones
`handlers.js`	Dispatches received events
`outbox.js`	Drains outbound writes with retry that cannot duplicate a record
`sync/documents.js`	Polls fee agreement state
`sync/transactions.js`	Nightly reconciliation of payments
`migrate/`	EspoCRM to GoHighLevel load, in ordered passes

6.2 The fee gate has two conditions, not one

GoHighLevel reports document status and payment status *independently*. A document can be signed and unpaid. `ghl.apply_fee_agreement()` is the only thing in the system that opens the gate, and it requires both: completed or accepted, **and** paid. A tag-based unlock gets this wrong.

6.3 Direction decides what happens to an inbound event

`core.property` is the system of record; GoHighLevel is downstream of it for listings. So an inbound record edit means somebody changed a property inside the CRM, against the grain of the architecture. Applying it would let the CRM silently overwrite the authoritative row. Dropping it would lose a real edit. It is neither: it goes to `ghl.review_queue` for a person to decide.

Events that genuinely originate in GoHighLevel — a signed document, a settled payment, a contact created by staff — are applied directly, because for those GoHighLevel is the source. Accepting a queued edit writes only an allowlist of band 1 columns, so a status change can never become a route to rewriting an address or a cost basis.

6.4 Nothing here has ever spoken to GoHighLevel

> **This is the most important caveat in the document, and it is easy to miss behind a passing test count. The integration is written and tested, but no line of it has made a request to GoHighLevel. Every test supplies a double: an injected `fetch`, a locally generated RSA keypair, a fake client object. Zero of the 63 tests reach the network.**

What that does and does not buy

Verified	Not verified
The logic: retry and backoff, deduplication, the two-condition fee gate, outbox resumability, migration ordering, allowlist enforcement, every privilege boundary.	Every assumption about how the real API actually behaves: response shapes, field names in practice, error bodies, and the webhook signature algorithm.
That the code does the right thing when GoHighLevel responds in the shape the code expects.	That GoHighLevel responds in that shape.

The assumption register

Rather than leave that as an unbounded worry, here is the finite list. Each row is a specific assumption the code makes, where it lives, and what happens if it is wrong. Checking these against one real account is a morning's work and closes the whole category.

#	Assumption	Where	If wrong
1	Signatures are PKCS#1 v1.5 over SHA-256, over the raw body	`signature.js`	Every delivery is rejected as forged. Loud, not silent.
2	Transaction list returns rows under `data` or `transactions`	`sync/transactions.js`	Sync silently reads zero rows and reports success. Quiet, and the worst of these.
3	Document list returns rows under `documents` with a `total`	`sync/documents.js`	The fee gate never opens for anyone. Loud once someone pays.
4	A document's payer is `recipients[0].contactId`	`sync/documents.js`	The document cannot be matched to a person; the gate stays shut.
5	Contact upsert returns `contact.id` or `id`	`outbox.js`	The id map is not written, so the association pass cannot resolve.
6	Record create returns `record.id` or `id`	`outbox.js`	Same, for properties.
7	Relation create takes `firstRecordId`/`secondRecordId`/`associationId`	`ghlClient.js`	Relations fail with a 422; the migration reports it rather than losing it.
8	Webhook payloads carry `type`, `webhookId` and `timestamp`	`webhookReceiver.js`	Events are recorded as `unknown` or rejected outright.
9	A 429 carries `Retry-After`	`ghlClient.js`	Backoff falls back to exponential. Harmless.

Assumption 2 is the one to check first. Most of these fail loudly — a rejected delivery, a 422, an empty gate somebody complains about. That one fails *quietly*: the sync reports success, the cursor advances, and the ledger simply never fills. A single real response settles it.

The code is written defensively against exactly this — the fallbacks above are why each row lists two candidate shapes rather than one — which lowers the odds but does not remove them. Guessing carefully is still guessing.

How to close it

Two artifacts, neither of which needs any code to be written: **one captured webhook delivery** (raw body plus its `x-wh-signature` header, which `worker/tools/capture-webhook.js` writes for you) and **one saved response from each of** `GET /payments/transactions` **and** `GET /proposals/document`. That is phase P0 in section 9, budgeted at one week, and it is first in the schedule for this reason.

7. The Marketplace

The browsing surface an investor actually uses: filters across the top, a map on the left, listing cards on the right, and a drill-down panel over the top. The layout is the one every property portal has converged on, for the reason they converged on it — price and geography are the two questions a buyer asks first, and the layout answers both at once.

7.1 What the browser can and cannot do

The front end contains no security logic. It does not decide whose address to show, which photographs to display, or which listings to draw. It draws what the API hands it, and it finds out that an address is withheld at the same moment the reader does — because `street_address` arrives as null.

This is worth stating because it is the difference between a gate and a curtain. A front end that receives every address and hides some of them is defeated by the browser's developer tools. This one is not sent them.

7.2 Searching

Control	Filters on	Where it runs
Price, beds, baths, square feet	Range on each	Bound parameters against `api.property`
Type, city	Exact match; the options are drawn from what this caller can see	Same query
Sort	Price, cap rate, size, bedrooms	An allowlisted ORDER BY, never interpolated
Plain English	Parsed into the same criteria as the controls above	See 7.5

Two properties of this that are structural rather than incidental. **A filter can only narrow.** The row policies and the view's masking run first; every filter clause runs inside that result, so there is no filter value that widens what a caller sees. And **filtering happens in the database**, so a listing the caller may not see is never sent and then hidden — it is not sent.

Even the facet ranges — the price floor and ceiling, the bedroom counts offered in the dropdown — are computed from the same policy-bounded relation. The bounds a caller sees are the bounds of their own data, so the filter bar itself does not leak the existence of listings they cannot open.

7.3 The map, and why gated pins are drawn as rings

Every listing carries true coordinates in band 2. An ungated viewer receives a deterministic offset of roughly a kilometre instead, seeded on the property id so repeated loads cannot be averaged back to the true point.

The map draws that difference rather than hiding it: an unlocked listing gets a sharp price pin, a gated one gets a soft ring around an approximate centre. Drawing a gated listing as a precise pin would claim an accuracy the data does not have, and would teach the viewer to trust a position that is deliberately wrong.

The map library and its tiles come from a CDN. When that CDN is unreachable — an air-gapped host, a restrictive proxy — a built-in renderer plots the same listings to scale with no dependencies, priced, clickable and hover-linked to their cards, labelled so nobody mistakes it for a basemap. A demo that shows a grey rectangle on a locked-down network is a demo that fails in the room it matters in.

7.4 The drill-down

One property, in full. The 404 for a listing the caller may not see is produced by the row policy, not by a check in application code: `api.property_detail` is a view over `api.property`, so an invisible property returns zero rows and the route reports not-found. Not *forbidden* — saying "forbidden" would confirm the listing exists to anyone who guessed an id.

Section	Contains	Band
Photography	Interior views. The front elevation is band 2 — see below	1, and 2
Headline figures	Cap rate, monthly rent, NOI, price per square foot	1
Income and expenses	Gross rent, vacancy, management, tax, insurance, maintenance, utilities, HOA, and the net	1
Area	Median household income, median price and rent, rent growth, vacancy, price-to-income, and this rent as a share of local income	1
The building	Lot, storeys, garage, parking, heating, cooling, roof year, renovations, features	1
Location	Street address, unit, exact coordinates, front elevation photograph	2

The whole financial analysis is band 1. That is a deliberate commercial choice, not an oversight: the analysis is what sells the listing, so withholding it would defeat the purpose of publishing at all. What is withheld is what identifies the parcel — nothing on the page is a teaser.

Photographs respect the same gate as the address. A picture of the front of a house identifies it as surely as its street number: a door number, a recognisable streetscape. So `core.property_media` carries `reveals_location`, and any image flagged that way is released on the same predicate as the address itself. Interiors stay public. Without this, the gate would be reopened through the picture gallery.

The expense breakdown sums exactly to the `opex_annual` published on the listing. That is enforced by generation rather than trusted: an investor who adds up the components gets the headline figure back, and a detail page whose arithmetic does not close is a page nobody underwrites on twice.

7.5 Plain-English search

A free-text box turns "3 bed duplex in Cleveland under 200k, best yield" into `{min_beds:3, property_type:'Duplex', city:'Cleveland', max_price:200000, sort:'cap_desc'}` and shows the reader what it understood, so the box is never opaque about what it did.

It is a rules parser today. It is not a language model and it does not call one. It is built now because the shape of the feature — free text in, a bounded criteria object out — is the part that has to be right, and it is worth having that seam built and tested before a model is put behind it.

The design is what makes a model safe to add later. The parser's only output is a criteria object whose keys are fixed and whose values the query builder binds, and every object passes a validator on the way out. So the blast radius of a wrong answer — from these rules today or from a model tomorrow — is a bad search, never a bad query. Swapping in a model means replacing one function body; the validator is not optional in that world, it is the thing that makes model output executable.

7.6 Favourites and saved searches

Feature	Stored in	Visibility
Favourites	`core.saved_property`	Owner only, plus staff. Saving is checked against the same predicate the row policy uses, so an investor cannot favourite a listing their own policy hides
Saved searches	`core.saved_search`	Owner only, with no staff override. What an investor is hunting for is their own business

Search criteria are stored as `jsonb` because the filter set will keep growing and each addition should not be a migration. The cost of `jsonb` is that it accepts anything — so the storable keys are constrained at the table. A saved search is replayed later, possibly by different code, and what cannot be stored cannot be replayed.

The favourites view joins `api.property`, not `core.property`. An address hidden in the grid must not appear because the listing was favourited; inheriting the mask rather than reimplementing it is what guarantees the two cannot drift apart.

7.7 Photography

Twenty-five photographs were supplied by the operator and are served from `web/public/assets`. Each of the twenty-five listings is paired with one, deterministically by listing reference, so a rebuild does not reshuffle them — a listing whose picture changes on every deploy looks broken even when nothing is. Any listing without a photograph still falls back to a generated illustration: a deterministic flat vector drawing seeded on the property id, visibly an illustration rather than a photo-imitation.

These are not photographs of these properties, and that single fact settles two questions that would otherwise be judgement calls. `reveals_location` is false, because the flag means *identifying*, not *exterior* — a photograph of a different house identifies nothing, and gating it would withhold something that protects nobody while teaching the next reader the wrong meaning. And every one is captioned as representative, because a stock exterior presented without qualification reads as a picture of the property, which is a small deception that compounds: an investor who drives past and finds a different house stops trusting the numbers too.

Each row carries two files. `url` is the supplied 1280px original, used for the lead image on a detail page. `thumb_url` is a 720px copy used for cards and the gallery strip; twenty-five originals on one listings page came to 9.4 MB against 2.1 MB of thumbnails, which is the whole reason the second column exists. It is null when there

is no smaller version, so every reader coalesces rather than assumes. The thumbnail carries no visibility of its own — it is the same picture, released or withheld with the row that owns it, so it is never a way around a gated image.

> **Where the photographs came from is not established. They are recorded in `gov.data_right` as `STOCK-PHOTOGRAPHY`, marked unreviewed. That is not a formality: a stock licence permitting commercial use without attribution, a licence requiring attribution, and a listing photograph belonging to a broker are three very different positions that look identical on disk, and the difference only surfaces in a demand letter.**

Replacing any of this is a URL change. `core.property_media` stores urls and nothing outside the renderer assumes where they are served from. The gate lives in the database and does not care where the bytes come from.

8. Where Listing Data Comes From

A property in this marketplace is also listed somewhere else — an MLS, a consumer portal, a wholesaler's sheet. It goes under contract, it sells, it is withdrawn, and nobody tells us. An investor who calls about a house that went pending three weeks ago is the single most expensive kind of stale data this system can hold.

The wrinkle that shapes the entire design: escrow fails. A property that went pending comes back to market perhaps one time in five. So this cannot be a one-way “mark it gone” job. It has to follow the source in both directions, and a design that only knows how to retire a listing will quietly bury the ones that come back.

8.1 Where the data can actually come from

Worth being blunt, because the obvious answer is not available.

Route	What it gives	What it costs
MLS via RESO Web API	The correct answer. A standardised feed with a closed status vocabulary, served by essentially every modern MLS	An MLS membership or approved vendor status, a signed data agreement, and a per-MLS contract. Coverage is per-market, so a national portfolio means several
Bridge Interactive (Zillow Group)	MLS data under Zillow Group's own platform	Same gate: manual approval, restricted to industry partners, brokerages and MLS organisations
Zillow's public API	Nothing. The legacy Web API (ZWSID) was retired; there is no self-serve endpoint for live listing inventory	—
Vendor property APIs (e.g. RentCast)	Self-serve, no membership. Property records, rent comps, and a derived listing status	A key and a subscription. Status is derivative rather than primary, which is why it is treated as advisory here
Scraping a consumer portal	Apparent completeness	Outside the portal's terms, and technically the weakest option — see 8.5
A person with a browser	Authoritative, immediate, and available today	Someone's time. For a portfolio of a few dozen listings this is a real answer, not a placeholder

The system is built so that which of these is used is a row in a table, not a rewrite. Every source implements one adapter interface and every answer goes through the same reconciler, so adding the MLS feed later changes no logic that is already tested.

8.2 Four tables and one rule

Table	Holds
`feed.listing_source`	Who is telling us, and how far they are trusted: authoritative or advisory, whether they may retire a listing, and how many agreeing checks a change needs
`feed.property_external`	Our property against their key, with `last_checked_at` and `last_seen_at` kept apart — “we looked” and “we found it” are different facts, and their difference is how a delisting is detected

Table	Holds
`feed.observation`	Append-only. What a source said, when, verbatim, alongside our reading of it. Never updated
`feed.status_change`	What we actually did about it, why, and whether a worker or a person did it

The rule: an observation never writes a status directly. It is recorded, then reconciled. That separation is what makes the history answerable after the fact — "why is this pending?" has an answer with a timestamp and a source — and what lets an unreliable source raise a flag without being allowed to act.

8.3 Vocabulary is data, not code

Every feed has its own words. RESO says *Active Under Contract*; one portal says *Pending*; another says *Contingent* or *Accepting backup offers*. All four mean the same thing here. `feed.status_map` holds the translation, so a term a portal invented last week is a row, not a release.

A term nobody has mapped is **recorded and flagged, never guessed**. An unmapped status appears on an operator worklist as "we saw something we do not understand" rather than silently defaulting to whatever seems closest.

8.4 What the nightly job is allowed to do

The sweep runs once a night, walks every enabled watch oldest-check-first so a run cut short has still done the most overdue work, and rate-limits per source. It decides nothing: it records what it saw and the database decides what it means.

Signal	What happens
The source reports a status	Recorded. Acted on once `confirm_after` checks agree — one reading of a feed is not evidence
The source reports Active on something we had pending	Acted on **immediately**. See below
The listing is absent from the feed	Counted. After enough consecutive absences the listing becomes *withdrawn* — not *sold*. Gone is not the same as sold, so we do not guess; withdrawn is the reversible one
The request failed	Ignored entirely. An outage is not a market emptying
An advisory source disagrees with us	A flag on the staff review queue. No status changes
A staff member changed the status by hand	The reconciler defers. A person who set it knows something the feed does not

Two asymmetries, both deliberate. An error is never read as an absence: an adapter that reports a timeout as "missing" would, over one bad night, walk the entire portfolio to withdrawn, so an adapter in doubt must report an error. And coming back to market is acted on the first sighting rather than the second: a failed escrow is a house that is saleable *now*, and every night it waits shown as unavailable is a night of lost enquiries. The two mistakes do not cost the same, so they are not treated the same.

8.5 Why the scraper is a named seam and not an implementation

It is wired in, it is registered as a source, and it returns “not implemented”. The reason is engineering before it is legal.

> **A scraper cannot reliably distinguish *this listing is gone* from *the page changed shape*. Both render as a missing selector. So its most common failure mode is indistinguishable from its most destructive signal, and a scraper allowed to retire listings will one day retire the whole portfolio in a single run — correctly, as far as it can tell.**

If one is ever built, two flags stay where they are: it remains *advisory*, so it raises flags instead of changing statuses, and it remains barred from retiring a listing. Those are columns on its source row, already set that way, so switching it on does not silently grant it authority.

8.6 Getting a listing into the system today

Route	How
MLS feed	Set `RESO_BASE_URL` and `RESO_TOKEN`, activate the source row. Listings arrive complete and stay current
By hand, in bulk	`worker/tools/import-listing.js` takes a JSON file of the details and merges it — only the keys present are written, so re-running with one corrected field does not blank the rest
By hand, one status	A staff member records what they saw; the next sweep picks it up and it goes through the same reconciler as any feed, so it is auditable and reversible

The Irvine listing in the seed data (108 Fairgrove, 92618) shows the intended posture. It is tracked — two watches, a real external key — and it is a **draft** with null sizes and zero prices, so it reaches nobody. The facts held are the ones in the source URL; the price, the room counts and the photographs are licensed content that has not been obtained. Inventing plausible numbers to fill the gaps would produce exactly the confident wrong record this schema exists to prevent. It is tracked before it is trusted.

9. Data Rights and Compliance

Every listing here is somebody else's data, held under some instrument. Whether it may be shown, to whom, and for how long is not a software question and cannot be answered by looking at the code — but it decides whether a listing may be published, so it is recorded where the publication decision is made: in the database, next to the row.

A licence recorded in a spreadsheet is a licence that gets breached the week somebody forgets the spreadsheet exists.

None of this is legal advice, and none of it claims to know what any particular agreement says. It is the shape that holds whatever the signed agreements say, plus a register of the regimes identified as applying. Every row carries a review status, and an unreviewed right is visibly unreviewed rather than quietly treated as settled.

9.1 Three questions per instrument

Question	Recorded as	Why it decides anything
Where does it apply?	`gov.data_right_territory`	A feed licensed for one market does not cover a property in another. Using it there is a breach even though the software works perfectly — which is exactly why it needs a mechanical check
What may be done with it?	`gov.data_right_use`	“Show this to a registered user” is not “publish this to the open web”, and neither is “export it in bulk” or “train a model on it”. Eight uses, each answered separately
Until when, and in return for what?	`gov.obligation`	Attribution wording, refresh cadence, removal within N hours of a delisting, deletion on termination. The ones with deadlines are computed rather than remembered

Two details in that table are doing more work than they look. **Silence is not permission**: a use with no row, or one marked *unclear*, is refused — and the intake tool writes every one of the eight uses out explicitly, so a reader can tell “we never asked” from “they said no”. And **rights are scoped**: a right over listing facts says nothing about the photographs. Listing photography usually belongs to the photographer or the listing broker rather than the seller, and conflating the two is a well-worn way for a portal to get sued.

9.2 What makes a right actually apply

Four conditions, and they are AND rather than a score. A right applies only if it is counsel-confirmed, unexpired, covers the property's territory, and grants the specific use being asked about.

`gov.may_use(property_id, use, scope)` is the single question the publication path asks. Every clause in it is a reason the answer is *no*, stated positively, so a future clause cannot accidentally widen it.

A right cannot mark itself confirmed. `record-data-right.js` only sets that status when a reviewer is named on the command line, because “somebody set a flag in a JSON file” and “a lawyer read the contract” must not look the same afterwards.

9.3 Advisory first, blocking later

The business operates today. A control that refused to publish anything without a confirmed right would take a working marketplace off the air over paperwork that has simply not been transcribed yet — and a control like that gets reverted, not fixed.

Mode	Behaviour
`advisory` (default)	Publication proceeds and warns. Every uncovered listing appears in `gov.uncovered_publication` with the reason. The whole gap is visible on day one and nothing breaks
`blocking`	Publication requires a confirmed right, and the standing invariant fails on any remaining gap. This is the go-live gate, and the invariant is what keeps it shut once flipped

Advisory is a recorded decision with a visible consequence, not the absence of one. It sits in `gov.policy` with who changed it and why.

9.4 Fair housing, enforced structurally

This is the one part of the register with teeth in the running application, and it deserves the space.

A marketplace that lets a user filter, or an algorithm rank, on a protected characteristic — or on a proxy for one — is steering, whether or not anyone intended it. **The Fair Housing Act does not require intent.** A recommendation engine is as capable of it as a dropdown, which is why the register exists before there is a recommendation engine.

The proxies are the hard part, and they are listed explicitly, because a system that blocks `race` and permits `percent_white_by_tract` has blocked nothing.

Registered proxy	Proxy for	Why
School rating	Race, national origin	Ratings track catchment demographics. Offering one as a ranking axis is steering by another name
Crime index	Race, national origin	Reported-crime indices measure policing intensity as much as risk
Area median income as a ranking	Race, national origin	The marketplace holds it, and legitimately — it is how rent is estimated. Offering it as an axis a buyer sorts on is a different act
Neighbourhood desirability score	Multiple	A composite is a laundered version of whatever went into it

Three controls, at three levels. The register is a table (`gov.prohibited_dimension`). The standing invariant fails if any listed dimension is ever exposed as a readable column. And **the web tier refuses to start** if its own filter allowlist intersects the register — the check runs against the database rather than living in a comment above the filter array, because the array is what a future feature edits and a rule that exists only in a comment survives exactly until somebody is in a hurry.

What this does not catch is a dimension added under an innocent name. That is what the register's `basis` column and code review are for. It catches the careless case, which is the common one.

9.5 The regimes on the register

Identified, with the trigger condition that makes each one apply, and — the column that makes this more than a wall poster — where the constraint is actually enforced. A regime with no control is a gap that is visible rather than assumed.

Regime	Applies when	Status here
Fair Housing Act, and state equivalents	Always	Enforced structurally, three ways — see 9.4
RESPA s.8 — referral fees	If compensated in connection with a federally related mortgage loan, including by referring business	**Unresolved and material.** See below
State real estate licensing	Per state where property is marketed	Unresolved. Turns on whether the platform's activity is brokerage
MLS participation, IDX and VOW rules	From the moment any MLS feed is connected	Machinery built and unused: attribution, refresh and removal SLAs are modelled and computed
Copyright in photographs and copy	Any media not authored in-house	Media provenance tracked separately from listing facts
CCPA/CPRA and state privacy statutes	Per consumer residency and thresholds	**Gap.** No subject-request mechanism exists
TCPA — SMS and calls	Any marketing text or autodialled call, including via GoHighLevel	**Gap.** Consent is not captured or evidenced here
CAN-SPAM	Any commercial email	Delegated to GoHighLevel; suppression state is not mirrored locally
ECOA / Regulation B	If credit is applied for, referred or influenced	Deferred. Applies the day lender matching is built
GLBA and the Safeguards Rule	If the business is a “financial institution”	Turns on the same facts as RESPA
PCI DSS	Any handling of cardholder data	Deferred. No payment integration exists; use a redirected processor when it does
ADA / web accessibility	Public-facing web content	Unaudited. The map has no non-visual equivalent
GDPR	Only if EU or UK data subjects	Not applicable today, registered so the trigger stays visible
Site terms and unauthorised access	If any portal is read programmatically	No scraper implemented — see 8.5

RESPA section 8 is the largest open legal question in this product, and it is not a technical one. The model is a $750 fee that unlocks property information and connects investors to agents and lenders. Whether that is compensation for services actually rendered or an unlawful referral fee turns on the fee's substance rather than its label. It needs counsel before launch, not after — and it should go to counsel together with the state licensing and GLBA questions, because all three turn on the same facts about what the business actually does.

TCPA is the highest-frequency risk, because it is the one an ordinary marketing decision can breach, statutory damages are per violation, and consent has to be provable. Nothing in this system currently captures or evidences it.

9.6 Filling the register in

The register currently holds the synthetic demonstration data, a deliberately empty instrument for the externally-tracked Irvine address, and the operator-supplied photograph of it. It does not yet hold the instruments the business actually operates under, because those live in signed paper that has to be transcribed by a person.

`docs/data-rights-intake.md` is the questionnaire for that — five questions per instrument, a worked example, and the loader. `SELECT * FROM api.governance_status` says where things stand at any moment.

To see	Query
The summary	`SELECT * FROM api.governance_status`
Published with no confirmed right	`SELECT * FROM gov.uncovered_publication`
The regimes, and which have no control	`SELECT * FROM api.compliance_register`
Per property: what is held, and why it does or does not apply	`SELECT * FROM api.data_rights`

10. Getting Properties In

Staff build an analysis workbook per property — one .xlsm holding the offer, the rents, the expenses and a twenty-year projection. Those numbers used to reach the marketplace by being retyped. This is the path that replaces that: load the file into staging, look at what arrived, then release the whole batch or the specific rows that passed review.

10.1 Two steps, and why

```
python3 tools/workbook-to-json.py *.xlsm > batch.json
node worker/tools/load-intake.js batch.json --note "August sourcing"
```

Reading .xlsm needs a spreadsheet library, and the worker image has no dependency beyond the Postgres driver. Rather than drag one in, the conversion happens wherever the file already is — a laptop, the host — and what crosses into the database is plain JSON. The middle format is the point: it can be opened and read before anything touches the database.

The reader takes the workbook's `Import` sheet, by label. It is the only sheet carrying the address, and its 77 labels were identical across the workbooks checked. The flatter `One Row` export has no address, so it cannot stand alone.

10.2 Staging holds both the file and our reading of it

Table	Holds
`intake.batch`	One file, one upload, one person, one moment
`intake.row`	One property. The verbatim payload and the parsed columns, side by side and never merged
`intake.zip_centroid`	The coordinate the workbook does not carry

Keeping the raw payload untouched matters more than it looks. When a released listing turns out to say something surprising, the only useful question is whether the spreadsheet said that or whether we mistranslated it — and that question has no answer if the import overwrote its own input. So it is written once, never edited, and every parsed column beside it is a claim that can be checked against it. The review screen shows it behind a link on every row.

A workbook has no coordinates, and `core.property` requires a point. A ZIP centroid is accurate to about a mile, which is exactly what an ungated viewer is shown anyway. A ZIP with no centroid is a blocking error rather than a silent (0,0): a listing quietly pinned to the middle of the wrong city is worse than one that refuses to load.

10.3 Two fields deliberately not mapped

The workbook carries *Schools Rating (scale 3-30)* and a composite FAVORABLE/INSUFFICIENT deal score partly derived from it. Both are already registered in `gov.prohibited_dimension` as fair-housing proxies — see 9.4.

They stay in the raw payload, because the file is not silently edited and staff underwriting may legitimately weigh schools. They never become a column. `api.security_invariants()` fails if either name appears in `core` or `api`,

so this is enforced rather than intended.

10.4 Validation blocks, or warns, and the difference is deliberate

Level	Examples	Effect
error	No address, no usable price or rent, no coordinate, an address already listed, the same address twice in one file	The row cannot be approved and cannot be released
warning	Unusual bedroom count or floor area, expenses meeting or exceeding gross rent, an expense total that does not reconcile with its own components	Shown to the reviewer. Does not block

A reviewer forced to clear every oddity before releasing anything stops reading the oddities, and the warnings are then worth nothing. That is the whole reason for the split.

Both copies of a duplicated address are flagged, not just the second. The system cannot know which one was intended, so it refuses them together and leaves the choice to a person.

> **The reconciliation check applies the management fee and vacancy allowance to the rent before comparing, rather than summing only tax, insurance and maintenance. An earlier version did the latter and flagged every correctly built workbook — and a check that fires on everything is one a reviewer learns to click past, which costs more than not having it.**

10.5 The review screen

`/admin.html`, staff only. Batches down the left, rows on the right, with price, rent, NOI and a cap rate computed here rather than trusted from the file.

Action	What it does
Approve selected	Marks rows reviewed. A row with a blocking error is refused, and the screen says how many of the selected rows actually changed
Reject selected	With an optional reason, kept against the row
Release selected	Creates the listings. Disabled unless something approved is selected, and it names the count
what the file said	The verbatim payload for that row

Nothing reaches the marketplace without a person releasing it. An invalid row cannot be approved — approving past a blocking error is how validation stops meaning anything — a pending row cannot be released, and “select all” means “all the releasable ones”, leaving blocked rows exactly where they are. A staging table that auto-promotes on a green validation is not a review queue; it is an import with extra steps.

The screen contains no permission logic. Every action is one of four `api` functions granted to staff alone, so an investor who calls them is refused by the database rather than by an if-statement in the page — which is also why the page settles the question by asking the server for the queue rather than reading a role name out of the session.

10.6 Release records provenance, and says when that is not enough

Releasing writes a `gov.property_provenance` row against the batch's data right. The workbooks are held under `SDI-WORKBOOK`, which is recorded **unreviewed** on purpose:

What arrives in the file	Whose it is
The financial modelling — offer, rents, expenses, projections	SDI's own work. Needs no external instrument
The property description	Verbatim MLS listing copy, written by the listing agent. The right to republish it has not been established

Because `gov.may_use()` honours only counsel-confirmed rights, releasing under it publishes with an advisory warning and lists the property in `gov.uncovered_publication`. **The review screen reports that back at the moment of release** rather than leaving it to be found in a report later — the register working on real data rather than demonstration data.

11. What Is Tested

Not a coverage percentage. These are the specific claims that are checked, and the attacks that are run against them.

Read this alongside 6.4. The database suites exercise a real PostgreSQL and prove what they claim. The worker suite proves its logic against test doubles, and proves nothing about GoHighLevel's or an MLS's actual behaviour.

Suite	Checks	Notable
`sql/05_tests.sql`	11	Five attacks: direct base-table read, another investor's saved list, saving an invisible property, a VOLATILE predicate side-channel, and the standing invariants
`sql/07_ghl_tests.sql`	7	A signed-but-unpaid document must not open the gate; replay must not move a signature timestamp
`sql/10_review_tests.sql`	7	A payload smuggling an address and a cost basis alongside a legitimate status change; only the allowlist is applied
`sql/14_pipeline_tests.sql`	9	One agent cannot read another's deal by naming its id; one investor cannot read another's stage history
`sql/23_listing_sync_tests.sql`	10	The whole listing lifecycle: under contract, failed escrow back to market, a feed outage, a genuine delisting, an advisory source, and a status term nobody has mapped
`sql/27_governance_tests.sql`	10	A data right built one failing condition at a time, and the same confirmed right refusing to cover a property one state away
`sql/29_intake_tests.sql`	9	Spreadsheet to listing: an invalid row cannot be approved, a pending row cannot be released, and “release ALL” releases only what was approved
`web/ (npm test)`	34	Password verification cost on the failure path, session revocation on password change, lockout after repeated failures, and that no application role can read a credential hash

Suite	Checks	Notable
`worker/ (npm test)`	70	Signature forgery, replay, oversized bodies, rate-limit handling, ambiguous-create duplication, migration resumability, and the sweep's discipline — an error is never an absence, an advisory source cannot act. All against doubles, see 6.4

Every suite that changes demo data restores it. That is not tidiness: a test that leaves a property pending, or an investor's fee agreement open, breaks the side-by-side contrast the whole demo rests on — and does it silently, one run later.

`api.security_invariants()` must always return zero rows. It catches the four changes that quietly dismantle the model: USAGE granted on `core`, an internal column exposed to a non-admin, RLS switched off on a protected table, or a view created without `security_invoker`. Wire it into CI and a nightly check.

12. What Is Not Built

Stated plainly so nothing here is mistaken for finished.

Missing	Consequence
A real listing feed	The MLS adapter is written against the RESO standard and has never run against a live feed. Until credentials exist, listing status is maintained by staff. See section 8
The portal scraper	Deliberately unimplemented. The source row, the trust flags and the review queue that would receive its output all exist — see 8.5 for why it stays that way
Document storage	The signed PDF artifact is not stored
Messaging and unified inbox	Not started
Audit trail on band 2 and 3 reads	Who viewed which address, and when, is not recorded
Co-investment matching	The pipeline exists; the matching engine does not
A language model behind the search box	The text parser is rules, not a model. The seam and the validator that would make model output safe to execute are built and tested — see 7.5
Real listing photography	Images are generated illustrations. Swapping them is a URL change; the gate does not depend on where they are served from
Password reset and email delivery	Authentication works, but a person who forgets a password needs staff to set a new one
EspoCRM field mapping	The load ordering and resumability are built and tested. The field-level mapping needs the live EspoCRM schema
A hardened deployment	The stack runs under Docker Compose and has been deployed on a VM. Nothing is internet-facing, and nothing should be until TLS, a reverse proxy and secret management are in place
The data-rights register, filled in	The machinery is built and the demo data is covered. The instruments the business actually operates under have not been transcribed — see 9.6 and `docs/data-rights-intake.md`

Missing	Consequence
Authorised delivery of gated media	Real photographs are served as static files under `web/public/`, so the database controls who is TOLD a url, not who can fetch it. Fine for generated illustrations, not for a location-revealing photograph. Needs an authorising route or signed, expiring urls
Consent capture for SMS and email	TCPA consent is neither captured nor evidenced here, and GoHighLevel's suppression state is not mirrored. This system cannot currently prove an opt-out was honoured
Privacy subject requests	No access, deletion or correction mechanism. One implementation satisfies CCPA and most state statutes at once
Uploading a workbook through the browser	The review screen reviews and releases; it does not accept a file. Loading is two commands at a shell — see 10.1
Geocoding	Coordinates come from a small ZIP centroid table, accurate to about a mile. A new market needs a row added before its listings will validate
Editing a staged row before release	A wrong figure means correcting the workbook and reloading. There is no in-place edit, deliberately: an edited row no longer matches the payload it is stored beside

13. Next Steps

Section 12 lists what is missing. This section says what each item needs, in what order, what has to be true before a phase can start, and how each one is tested. Durations are working weeks for one experienced developer and are estimates, not commitments.

13.1 What each item needs, and how long

Ordered as recommended in 13.2. Estimates are working weeks for one experienced developer who already knows this codebase, and cover build plus the tests in 13.4. They exclude design iteration, review latency, and waiting on a third party — the three things that actually move dates.

#	Item	What it is	Needs before starting	Est.
P0	Signature verification and first deployment	Confirm how GoHighLevel signs webhooks, and deploy the receiver that checks it.	A public HTTPS host and a GoHighLevel account. Nothing else in the system.	1w
P1	Authentication and sessions	Real sign-in. A session resolves to a person id and a role, which the web tier assumes for the transaction.	A decision on identity provider: own accounts, or an external one. No database change.	3w
P2	EspoCRM field mapping and rehearsal	Which EspoCRM field becomes which column, and which of the 30+ SDI metrics are inputs rather than derived.	Read access to live EspoCRM or a full export. Load ordering already built and tested.	4w
P3	Audit trail on gated reads	A record of who saw which address, and when.	P1. A decision on where band 2 is released from — PostgreSQL cannot trigger on SELECT.	2w
P4	Public marketplace UI	The real browsing surface: search, filters, property cards, masked map.	P1 for the gated half, P2 for real content, and design direction.	4w
P5	Investor portal and fee flow	Registration, the fee agreement, and the unlock.	P4, and a payment provider connected in GoHighLevel. The CRM side is already built.	3w
P6	Document storage	The signed PDF kept as an artifact, not only as a status flag.	A storage and retention decision. The signed document is a financial record.	2w
P7	Agent portal	An agent's own assignments and conversations, and nothing else.	P1. Per-agent isolation is already enforced and tested at the database level.	2w
P8	Messaging and unified inbox	Email, SMS and WhatsApp threaded against a contact and a property.	A decision on whether GoHighLevel owns the conversation or only relays it. That sets the data model, so settle it first.	3w
P9	External status feed	Nightly check that a listing has not gone pending or sold elsewhere.	A source. A licensed feed is strongly preferable to scraping a portal that forbids it.	2w

#	Item	What it is	Needs before starting	Est.
P10	Co-investment matching	Pairing two vetted investors on one property.	The matching rules, in writing. The COINVEST pipeline and stages already exist.	3w
P11	Hardening and cutover	Production configuration, backups, restore rehearsal, go-live.	Everything above that is in scope for launch.	2w

Total build effort is roughly **31 developer-weeks**. The calendar in 13.2 is 20 weeks because P2, P7 and P9 run alongside other work rather than after it. With one developer and no parallelism the same scope is about 31 weeks; the difference is entirely whether the EspoCRM and operations tracks can proceed independently.

13.2 Recommended sequence

Two things drive this order. **Authentication gates everything user-facing**, so it goes first and almost nothing can be demonstrated to a real user before it lands. And **the audit trail should precede real investor data**, not follow it — the first time anyone asks who saw an address, the answer has to already exist.

P0 is deliberately tiny and first. It deploys nothing but a webhook receiver, which authenticates by signature rather than by session, so it needs no login and exposes no data. It settles the one unverified item in the system and proves the deployment path at the same time.

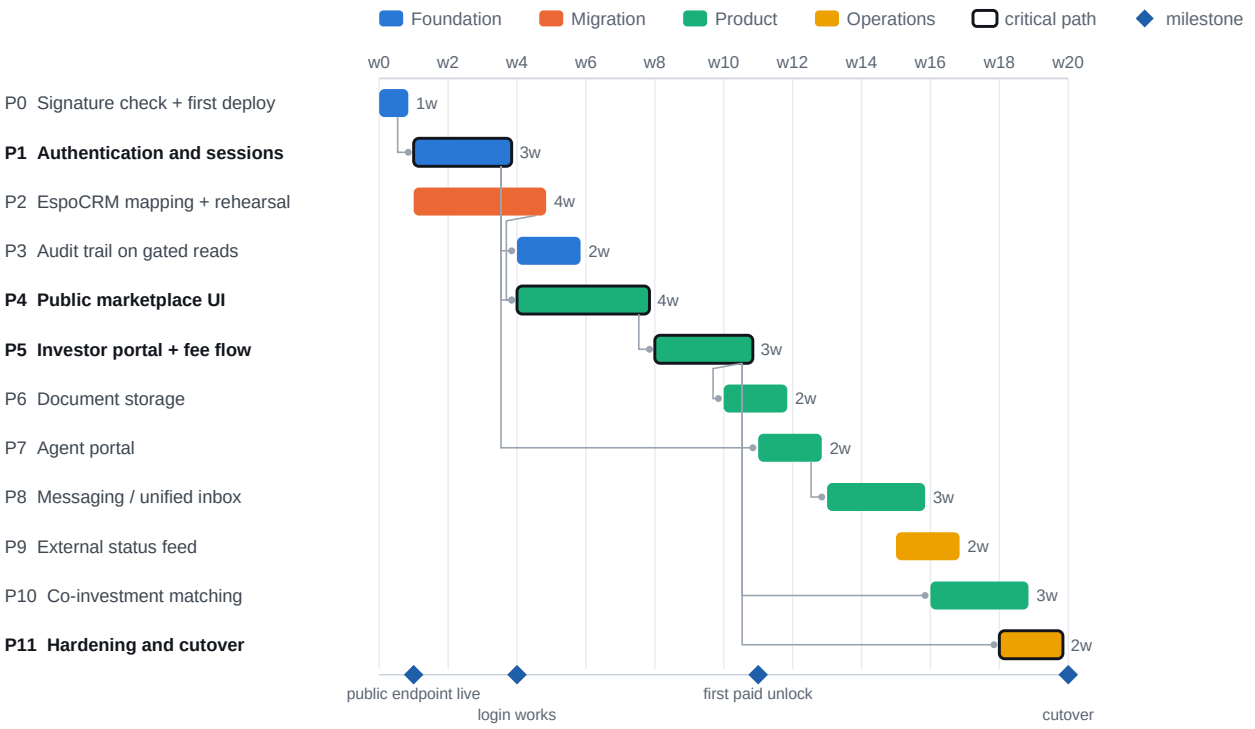


Figure 1. Indicative schedule, one developer. P2 runs alongside P1 because the migration work needs EspoCRM access rather than the new authentication, so the two do not contend.

The critical path is P1 → P4 → P5: authentication, then the browsing surface, then the paid unlock. Everything else can move without moving the launch date. If the schedule has to compress, P8, P9 and P10 are the ones to defer — none of them is required for an investor to find a property, pay, and see the address.

13.3 What must be true before a phase starts

Phase	Entry condition	Done when
P0	A host with a public HTTPS address and a GoHighLevel account	A live delivery verifies against the published key, and the algorithm is confirmed in code and in the spec document
P1	A decision on the identity provider: own accounts, or an external one	A session resolves to a person id and role; every existing SQL walkthrough still passes unchanged

Phase	Entry condition	Done when
P2	Read access to live EspoCRM, plus the field mapping agreed in writing	A full rehearsal load into a disposable sub-account reconciles with zero shortfall and zero unresolved links
P3	P1 complete. A decision on where band 2 is released from	Every band 2 and band 3 read is attributable to a person and a time; the attack tests still pass
P4	P1 complete; design direction; listing content from P2	An anonymous visitor can browse and filter, and cannot obtain an address by any route including the network tab
P5	P4 complete; a payment provider connected in GoHighLevel	An investor signs, pays, and the address unlocks — driven end to end against a GoHighLevel test sub-account
P6	A storage and retention decision	The signed PDF is retrievable and its retention is enforced
P7	P1 complete	An agent sees only their own assignments and conversations, proven by test, not by inspection
P8	The ownership decision in 13.1	A thread is readable against both the contact and the property
P9	A data source	A status change reaches the review queue and no listing changes without a human
P10	Matching rules in writing	Two vetted investors are matched and both are notified
P11	Everything above that is in scope for launch	Cutover rehearsed on staging, with a tested rollback

13.4 How each phase is tested

Every phase adds its own tests and must leave the existing ones green. That second half is the part that usually erodes, so it is stated as a gate rather than a habit.

The standing regression suite

Suite	Checks	Must remain
`sql/05_tests.sql`	11	All five attacks refused
`sql/07_ghl_tests.sql`	7	Signed-but-unpaid still does not open the gate
`sql/10_review_tests.sql`	7	The allowlist still refuses band 2 and band 3 columns
`sql/14_pipeline_tests.sql`	9	No cross-agent or cross-investor read
`worker/ npm test`	63	All passing
`api.security_invariants()`	—	Zero rows, on every build

Run the whole thing with `./run.sh`, which loads the schema, runs all four walkthroughs and the worker suite in one pass. Wire `api.security_invariants()` into CI and into a nightly job: it catches the four changes that quietly dismantle the model, and it catches them whether they came from a migration, a hotfix, or a well-meant grant.

Per-phase development tests

Phase	New tests it must bring
P0	Signature verification against a captured live delivery, added as a fixture so it is checked forever after
P1	Session to role mapping; an expired session; a tampered session; a session for a deactivated person
P2	Reconciliation counts; a resumed load after a forced failure; a link whose endpoints are missing
P3	An audit row exists for every band 2 and band 3 release; the audit log cannot be written by the reader
P4	An anonymous HTTP response contains no restricted field — asserted on the response body, not the rendered page
P5	Signed-and-paid unlocks; signed-and-unpaid does not; a refund after unlock
P6	The stored artifact matches what was signed; retention removes it on schedule
P7	One agent cannot reach another's assignment or conversation by id
P8	A message is attributable to a contact and a property; no cross-tenant leak
P9	A source change raises exactly one review item; a flapping source does not raise many
P10	A match requires two vetted parties; an unvetted party is never matched
P11	A restore from backup produces a working system; rollback returns to the prior version

Two habits worth keeping, both learned building what already exists. Write the test that tries the attack, not only the one that proves the feature — several of the checks in the suite exist because the naive version of a test passed while

proving nothing. And re-run the whole suite, not the file you are working on: two of the bugs found so far only appeared when suites ran together.

13.5 Validation methods to consider

Section 13.4 says what to test. This says *how*, and which technique earns its place where. Not all of these are worth adopting; they are listed with the judgement attached rather than as a checklist to complete.

Method	What it is good for here	Worth it?
Integration tests against a real PostgreSQL	Row-level security, policies and grants cannot be mocked — a mock will happily return rows the real database would refuse. The existing suite already works this way.	Already in use. Keep it; never substitute mocks for the database.
Adversarial tests	A test that tries the attack, not one that proves the feature. Every 'ATTACK' check in the SQL walkthroughs is one, and several exist because the naive version passed while proving nothing.	Essential. One per privilege boundary, minimum.
Property-based testing	Generate random (viewer, property, entitlement) triples and assert the invariant: a viewer without a settled agreement never receives a street address. This explores combinations nobody thought to write down.	High value from P3 onward. The visibility model is an unusually good fit.
Contract testing against the OpenAPI specs	Validate outbound request shapes against GoHighLevel's published specification, so a field rename is caught at build time rather than as a 422 in production.	Worth it once P2 volume is real.
Fault injection	Deliberately fail mid-operation: kill the worker between an API call and its database write, redeliver a webhook, duplicate an outbox row. The resumability claims are only true if they survive this.	Essential before P11. Cheap to run, and the failure it catches is data corruption.
Load testing	The GoHighLevel burst limit is 100 requests per 10 seconds shared across all callers. Confirm the cached read model absorbs concurrent browsing rather than passing it through.	Before P4 goes public. Model realistic concurrency, not a synthetic peak.
Penetration testing	An independent attempt to obtain a street address without paying. The commercial model rests on that being impossible, which makes it the one thing worth an outside opinion.	Before go-live. Scope it explicitly at the gate.
Migration reconciliation	Counts and spot checks between source and destination after every rehearsal, with unresolved links reported rather than dropped.	Already built. Run it on every rehearsal, not just the final one.
User acceptance testing	Walk the seeded personas through real tasks with real staff. Ruth and Marcus exist precisely so the difference is demonstrable to a non-technical observer.	Per phase, from P4.
Accessibility testing	The public marketplace is a consumer surface. Keyboard navigation, contrast and screen reader behaviour on listing cards and filters.	During P4, not after. Retrofitting is far more expensive.

Method	What it is good for here	Worth it?
Data quality checks	Standing assertions on the ledger: no negative amounts, no refund exceeding its charge, no transaction pointing at a missing invoice. Some are already constraints.	Promote to constraints where possible — a constraint cannot be forgotten.
Static analysis and dependency audit	Lint, type checking, and a scan of the dependency tree. This project has one runtime dependency, which keeps this cheap.	In CI from now. The cost is near zero while the tree is small.

Three rules to attach to whichever of these you adopt

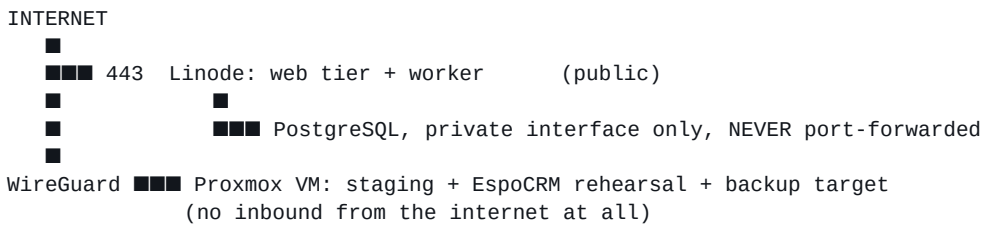
- **Run the whole suite, not the file you are working on.** Two of the bugs found while building this only appeared when suites ran together — one because two test files shared a database and raced.
- **A test that cannot fail is not a test.** Two checks here originally passed for the wrong reason: a permission error fired before the code under test was ever reached. Make each new test fail once, deliberately, before trusting it.
- **Wire `api.security_invariants()` into CI and a nightly job.** It is the only check that catches a policy quietly dismantled by a later migration, and it costs one query.

13.6 Deployment

Both available options are suitable, for different jobs. The recommendation is to use both rather than choose.

Environment	Role	Why
Proxmox VM (local)	Development and staging. The full stack, no public exposure.	The EspoCRM rehearsal belongs here: real client data never leaves the network, and a rehearsal against a snapshot is repeatable in a way a live read is not. A VM rather than a container — PostgreSQL wants a stable filesystem and its own kernel-level tuning.
Linode (public)	Production. Web tier and worker, publicly reachable.	P0 needs a public HTTPS endpoint for GoHighLevel to deliver to, and nothing else does until P4. Start with the smallest instance that runs the worker.

Topology



Four rules for the production environment, each of which is cheap now and expensive to retrofit:

- **PostgreSQL is never exposed to the internet.** Bind it to the private interface. The proxy this project was built behind refuses raw database ports for exactly this reason.
- `sdi_test_admin` **must not exist in production.** It carries BYPASSRLS and exists only for test fixtures. `worker/test/bootstrap.sql` is not a deployment script.
- `sql/99_local_logins.sql` **must not be loaded in production.** Its passwords are published in a public repository. Production roles get credentials from the deployment, not from the repository.
- **The GoHighLevel token lives in the environment**, never in a file and never in browser-reachable code. It is scoped to an entire sub-account.

Backups are the one thing to set up before there is anything worth backing up, because that is the only time it is easy. Nightly `pg_dump` to the Proxmox side over WireGuard, and a restore rehearsed at least once — an untested backup is a belief, not a backup.

14. Where Things Are

Path	Contents
`sql/01–04`	Schema, RLS policies, masking views, demo data
`sql/05, 07, 10, 14, 23, 27, 29`	The seven walkthroughs. Each is readable top to bottom as an argument, and each restores what it changes
`sql/06, 08, 09`	GoHighLevel bridge, review queue, review actions
`sql/11–13`	Deals, stage history, pipeline policies and seed
`sql/15_auth.sql`	Credentials and sessions — see 3.1
`sql/16–17, 20`	The demo dataset: 24 listings, their passwords, and the generated detail and market data
`sql/18–19`	Property detail, market areas, photographs; saved searches
`sql/21–22`	Listing sources, status vocabulary, reconciliation — see section 8
`sql/24–26`	Data rights, territories, permitted uses, the compliance register, the fair-housing prohibited list — see section 9
`sql/28`	Spreadsheet intake: staging, validation, review and release — see section 10
`tools/`	The workbook reader. Python, because it needs a spreadsheet library the worker deliberately does not
`sql/99_local_logins.sql`	Demo passwords. Local development only
`web/`	The marketplace. `server.js`, `auth.js`, `nlq.js` (the text parser), `media.js` (the illustrations), and `public/` (the marketplace and `admin.html`, the review screen)
`web/test/`	34 tests
`worker/src/`	The GoHighLevel worker and the listing sweep (`listings/`)
`worker/test/`	69 tests
`worker/tools/`	Webhook capture, the nightly sweep, the listing importer, the data-rights loader and the intake loader
`docs/data-rights-intake.md`	The questionnaire that turns signed agreements into enforceable rows
`docs/`	This document and the GoHighLevel interface specification, with their generators
`README.md`	The same ground in more technical detail, including the reasoning behind each design decision

There is no INSTALL file; section 2 of this document and the README's opening section cover installation. Both PDFs in `docs/` are generated by committed scripts rather than written by hand, so they can be regenerated as the system changes and cannot quietly drift out of date.

15. Build and Release

How the three container images are produced, how a release is cut, and what a deployment actually pulls. The short version: nothing is built on a host. Every image is built by CI from a tagged commit and pulled by tag — which is what makes the stack deployable to Swarm, which cannot build at all, and keeps build tooling off the machine serving traffic.

15.1 What gets built

Image	Base	Built from	Contains
`db`	`postgres:16`	`docker/db.Dockerfile`	Every schema file, copied into `/docker-entrypoint-initdb.d/`, plus the role-login script. The test walkthroughs are deliberately NOT copied
`web`	`node:22-alpine`	`web/Dockerfile`	`server.js`, `auth.js`, `media.js`, `nlq.js` and `public/`
`worker`	`node:22-alpine`	`worker/Dockerfile`	`src/` and `tools/` — the operator entry points have to be inside the image, not merely in the repository

One runtime dependency across both Node images: `pg`. Installed with `npm ci` rather than `npm install`, so the build is reproducible and fails loudly if the lockfile and the manifest disagree. Both run as the unprivileged `node` user.

> **The `web` image names each module it copies rather than globbing. That is not fussiness: an earlier version listed only two of the four, which produced an image that passed every test and crashed on its first require — a break invisible outside a container build.**

15.2 Cutting a release

```
# 1. update the release level and its entry
echo 0.10.0 > VERSION
$EDITOR CHANGELOG.md
python3 docs/extract_schema.py            # if the schema changed
python3 docs/generate_system_documentation.py
python3 docs/generate_test_plan.py

# 2. commit, tag, push
git commit -am "Release 0.10.0"
git tag -a v0.10.0 -m "0.10.0"
git push && git push --tags
```

The push builds three images; the tag builds them again carrying the version. `VERSION` and `CHANGELOG.md` are read by this document at generation time, so the release level on the cover and the change log in Appendix B cannot disagree with the repository.

15.3 What CI produces

Tag	When	Use it for
`latest`	Every push to `main` or the working branch	The demo. It moves

Tag	When	Use it for
`v0.10.0`, `0.10`	A `v*` tag	Anything real. Pin it
`sha-<full sha>`	Every build	Reproducing an exact deployment
Branch name	Every branch build	Testing a branch before it merges

`docker-compose.release.yml` reads `SDI_VERSION` and defaults to `latest`. Set it to a release tag for anything that matters:

```
SDI_VERSION=v0.9.13 docker compose -f docker-compose.release.yml up -d
```

The workflow pins `latest` to two named branches rather than using the `is_default_branch` template. This repository's default branch is a working branch, so that template evaluated false and the first release published only an immutable `sha-` tag — leaving nothing for a compose file referencing `latest` to pull, which looked exactly like a permissions problem and was not.

15.4 Building locally

Only needed to test a Dockerfile change; the deployment never does this.

```
docker compose build                    # all three, from source
docker compose up -d

./db-rebuild.sh                         # database only, no Docker
(cd web && npm test) && (cd worker && npm test)
```

Script	Does
`run.sh`	Loads every schema file into a local PostgreSQL, then runs all seven walkthroughs
`db-rebuild.sh`	Drops and reloads the database, including the test fixture role that is easy to forget
`docs/extract_schema.py`	Snapshots the live schema to `docs/schema-snapshot.json` for Appendix A

Adding a schema file means editing three lists, and missing one produces a failure far from its cause: `run.sh` for local development, `db-rebuild.sh` for a fast rebuild, and `docker/db.Dockerfile` for the image. Test walkthroughs go in the first two only.

15.5 Release checklist

#	Check	How
1	Every test passes	`npm test` in `web/` and `worker/`; the seven SQL walkthroughs
2	No standing invariant is violated	`SELECT * FROM api.security_invariants()` returns zero rows
3	A clean rebuild works	`./db-rebuild.sh` from nothing, not an incremental patch
4	The demo fixture is intact	Marcus's gate shut, Ruth's open. A test suite that leaves it otherwise has broken the demo
5	New schema files are in all three lists	See 15.4
6	The schema snapshot is current	`python3 docs/extract_schema.py`
7	Both PDFs regenerated	They read `VERSION` and `CHANGELOG.md` at generation time

#	Check	How
8	The change log entry is written	`CHANGELOG.md`, before the tag

Appendix A. Table Definitions

Every column of every base table, read from the live database by `docs/extract_schema.py` and stored in `docs/schema-snapshot.json`. Regenerate the snapshot after any schema change and rebuild this document; the appendix cannot then describe a schema the system does not have.

Marker	Meaning
PK	Part of the primary key
FK	Foreign key; the referenced table is named in the same row
UQ	Covered by a unique constraint
CK	Covered by a check constraint
not null	The column is `NOT NULL`

Schema `core`

`core.brand`

4 columns · row-level security enforced

Column	Type	Null	Key	Default / references
`brand_code`	`text`	not null	PK	
`display_name`	`text`	not null		
`service_tier`	`text`	not null	CK	
`platform_fee`	`numeric(10,2)`	not null		

`core.credential`

6 columns · row-level security enforced · No RLS policy exists on purpose. Direct access is refused for every role; api.authenticate() and api.set_password() are the only paths.

Column	Type	Null	Key	Default / references
`person_id`	`uuid`	not null	FK PK	→ core.person
`password_hash`	`text`	not null		
`failed_attempts`	`integer`	not null		0
`locked_until`	`timestamptz`			
`password_set_at`	`timestamptz`	not null		now()
`updated_at`	`timestamptz`	not null		now()

`core.deal`

13 columns · row-level security enforced

Column	Type	Null	Key	Default / references
deal_id	uuid	not null	PK	gen_random_uuid()
external_ref	text		UQ	
property_id	uuid	not null	FK	→ core.property
investor_id	uuid		FK	→ core.person
agent_id	uuid		FK	→ core.person
pipeline_code	text	not null	FK	→ core.pipeline_stage
stage_code	text	not null	FK	→ core.pipeline_stage
amount	numeric(12,2)			
opened_at	timestamptz	not null	CK	now()
closed_at	timestamptz		CK	
lost_reason	text			
created_at	timestamptz	not null		now()
updated_at	timestamptz	not null		now()

core.deal_stage_history

7 columns · row-level security enforced · Append-only, written by trigger. The application cannot forget to log a transition, and cannot rewrite one after the fact.

Column	Type	Null	Key	Default / references
id	bigint	not null	PK	auto-increment
deal_id	uuid	not null	FK	→ core.deal
from_stage	text			
to_stage	text	not null		
changed_at	timestamptz	not null		now()
changed_by	uuid			
seconds_in_from	bigint			

core.market_area

11 columns · row-level security enforced

Column	Type	Null	Key	Default / references
area_id	text	not null	PK	
city	text	not null	UQ	

Column	Type	Null	Key	Default / references
state	char(2)	not null	UQ	
median_household_income	numeric(12,2)			
median_home_price	numeric(12,2)			
median_rent_monthly	numeric(10,2)			
rent_growth_1y_bps	integer			
vacancy_rate_bps	integer			
population	integer			
price_to_income	numeric(5,2)			
as_of	date	not null		CURRENT_DATE

core.media_event

7 columns · row-level security enforced

Column	Type	Null	Key	Default / references
event_id	bigint	not null	PK	auto-increment
media_id	uuid	not null		
property_id	uuid			
action	text	not null	CK	
actor_id	uuid			
detail	jsonb	not null		'{}'
at	timestamptz	not null		now()

core.person

7 columns · row-level security enforced

Column	Type	Null	Key	Default / references
person_id	uuid	not null	PK	gen_random_uuid()
role	actor_role	not null		
full_name	text	not null		
email	text	not null	UQ	
fee_agreement_signed_at	timestamptz			
home_brand	text		FK	→ core.brand

Column	Type	Null	Key	Default / references
active	boolean	not null		true

core.pipeline

4 columns · row-level security enforced

Column	Type	Null	Key	Default / references
pipeline_code	text	not null	PK	
display_name	text	not null		
brand_code	text		FK	→ core.brand
active	boolean	not null		true

core.pipeline_stage

6 columns · row-level security enforced

Column	Type	Null	Key	Default / references
pipeline_code	text	not null	FK PK UQ	→ core.pipeline
stage_code	text	not null	PK	
display_name	text	not null		
position	integer	not null	UQ	
is_won	boolean	not null	CK	false
is_lost	boolean	not null	CK	false

core.property

27 columns · row-level security enforced

Column	Type	Null	Key	Default / references
property_id	uuid	not null	PK	gen_random_uuid()
listing_ref	text	not null	UQ	
status	text	not null	CK	
city	text	not null		
state	char(2)	not null		
zip	text	not null		
property_type	text	not null		
beds	smallint			

Column	Type	Null	Key	Default / references
baths	numeric(3,1)			
sqft	integer			
year_built	smallint			
list_price	numeric(12,2)	not null		
gross_rent_annual	numeric(12,2)	not null		
opex_annual	numeric(12,2)	not null		
hoa_annual	numeric(12,2)	not null		0
noi_annual	numeric(12,2)			
cap_rate	numeric(6,4)			
street_address	text	not null		
unit	text			
lat	numeric(9,6)	not null		
lng	numeric(9,6)	not null		
parcel_number	text			
seller_disclosure	text			
acquisition_cost	numeric(12,2)			
source_channel	text			
internal_notes	text			
created_at	timestamptz	not null		now()

core.property_assignment

3 columns · row-level security enforced

Column	Type	Null	Key	Default / references
property_id	uuid	not null	FK PK	→ core.property
person_id	uuid	not null	FK PK	→ core.person
assign_role	text	not null	CK PK	

core.property_brand

4 columns · row-level security enforced

Column	Type	Null	Key	Default / references
property_id	uuid	not null	FK PK	→ core.property

Column	Type	Null	Key	Default / references
brand_code	text	not null	FK PK	→ core.brand
published	boolean	not null		false
brand_price	numeric(12,2)			

core.property_detail

22 columns · row-level security enforced

Column	Type	Null	Key	Default / references
property_id	uuid	not null	FK PK	→ core.property
headline	text			
description	text			
market_rent_monthly	numeric(10,2)			
rent_basis	text		CK	
property_tax_annual	numeric(10,2)			
insurance_annual	numeric(10,2)			
utilities_monthly	numeric(10,2)			
utilities_paid_by	text		CK	
maintenance_annual	numeric(10,2)			
management_fee_bps	integer			
vacancy_allowance_bps	integer			
lot_sqft	integer			
stories	smallint			
garage_spaces	smallint			
heating	text			
cooling	text			
roof_year	smallint			
last_renovated	smallint			
parking	text			
features	jsonb	not null		'[]'
updated_at	timestamptz	not null		now()

core.property_media

22 columns · row-level security enforced

Column	Type	Null	Key	Default / references
media_id	uuid	not null	PK	gen_random_uuid()
property_id	uuid	not null	FK	→ core.property
url	text		CK	
thumb_url	text			
caption	text			
position	integer	not null		0
is_primary	boolean	not null		false
reveals_location	boolean	not null		false
storage_path	text		CK	
thumb_path	text			
content_sha256	bytea			
byte_size	integer			
width	integer			
height	integer			
original_name	text			
state	media_state	not null		'published'
published_at	timestamptz			
deleted_at	timestamptz			
purge_after	date			
legal_hold	boolean	not null		false
created_by	uuid		FK	→ core.person
created_at	timestamptz	not null		now()

core.saved_property

3 columns · row-level security enforced

Column	Type	Null	Key	Default / references
person_id	uuid	not null	FK PK	→ core.person
property_id	uuid	not null	FK PK	→ core.property
saved_at	timestamptz	not null		now()

core.saved_search

7 columns · row-level security enforced

Column	Type	Null	Key	Default / references
search_id	uuid	not null	PK	gen_random_uuid()
person_id	uuid	not null	FK UQ	→ core.person
name	text	not null	CK UQ	
criteria	jsonb	not null	CK	'{}'
created_at	timestamptz	not null		now()
last_run_at	timestamptz			
run_count	integer	not null		0

core.session

8 columns · row-level security enforced

Column	Type	Null	Key	Default / references
token_hash	bytea	not null	PK	
person_id	uuid	not null	FK	→ core.person
created_at	timestamptz	not null		now()
expires_at	timestamptz	not null		
last_seen_at	timestamptz	not null		now()
revoked_at	timestamptz			
user_agent	text			
ip	inet			

Schema feed

feed.listing_source

11 columns · no row-level security

Column	Type	Null	Key	Default / references
source_code	text	not null	PK	
name	text	not null		
kind	text	not null	CK	
base_url	text			
authoritative	boolean	not null		false

Column	Type	Null	Key	Default / references
confirm_after	smallint	not null	CK	2
may_retire	boolean	not null		false
active	boolean	not null		true
check_cron	text			
notes	text			
created_at	timestamptz	not null		now()

feed.manual_check

8 columns · no row-level security

Column	Type	Null	Key	Default / references
check_id	bigint	not null	PK	auto-increment
property_id	uuid	not null	FK	→ core.property
raw_status	text	not null		
list_price	numeric(12,2)			
note	text			
noted_by	uuid		FK	→ core.person
noted_at	timestamptz	not null		now()
consumed_at	timestamptz			

feed.observation

11 columns · no row-level security

Column	Type	Null	Key	Default / references
observation_id	bigint	not null	PK	auto-increment
run_id	uuid			
property_id	uuid	not null	FK	→ core.property
source_code	text	not null	FK	→ feed.listing_source
observed_at	timestamptz	not null		now()
outcome	text	not null	CK	
raw_status	text			
mapped_status	text			
list_price	numeric(12,2)			

Column	Type	Null	Key	Default / references
payload	jsonb			
error_detail	text			

feed.property_external

12 columns · no row-level security

Column	Type	Null	Key	Default / references
property_id	uuid	not null	FK PK	→ core.property
source_code	text	not null	FK PK UQ	→ feed.listing_source
external_id	text	not null	UQ	
external_url	text			
first_seen_at	timestamptz	not null		now()
last_checked_at	timestamptz			
last_seen_at	timestamptz			
last_raw_status	text			
last_price	numeric(12,2)			
miss_streak	smallint	not null		0
agree_streak	smallint	not null		0
enabled	boolean	not null		true

feed.review_flag

10 columns · no row-level security

Column	Type	Null	Key	Default / references
flag_id	bigint	not null	PK	auto-increment
property_id	uuid	not null	FK	→ core.property
source_code	text		FK	→ feed.listing_source
observation_id	bigint		FK	→ feed.observation
kind	text	not null		
detail	text	not null		
raised_at	timestamptz	not null		now()
resolved_at	timestamptz			

Column	Type	Null	Key	Default / references
resolved_by	uuid		FK	→ core.person
resolution	text			

feed.status_change

9 columns · no row-level security

Column	Type	Null	Key	Default / references
change_id	bigint	not null	PK	auto-increment
property_id	uuid	not null	FK	→ core.property
from_status	text	not null		
to_status	text	not null		
reason	text	not null		
source_code	text		FK	→ feed.listing_source
observation_id	bigint		FK	→ feed.observation
actor	text	not null		'worker'
changed_at	timestamptz	not null		now()

feed.status_map

3 columns · no row-level security

Column	Type	Null	Key	Default / references
source_code	text	not null	FK PK	→ feed.listing_source
raw_status	text	not null	PK	
mapped_status	text		CK	

Schema ghl

ghl.fee_agreement

10 columns · no row-level security

Column	Type	Null	Key	Default / references
document_id	text	not null	PK	
location_id	text	not null		
person_id	uuid		FK	→ core.person
ghl_contact_id	text			

Column	Type	Null	Key	Default / references
status	text	not null		
payment_status	text	not null		
grand_total	numeric(12,2)			
ghl_updated_at	timestamptz	not null		
observed_at	timestamptz	not null		now()
raw	jsonb	not null		

ghl.id_map

6 columns · no row-level security

Column	Type	Null	Key	Default / references
entity_type	text	not null	PK	
local_id	uuid	not null	PK	
ghl_id	text	not null	UQ	
ghl_object	object_kind	not null		
location_id	text	not null	PK UQ	
synced_at	timestamptz	not null		now()

ghl.outbox

13 columns · no row-level security

Column	Type	Null	Key	Default / references
id	bigint	not null	PK	auto-increment
idempotency_key	text	not null	UQ	
operation	text	not null		
entity_type	text			
local_id	uuid			
payload	jsonb	not null		
state	outbox_state	not null		'pending'
attempts	integer	not null		0
next_attempt_at	timestamptz	not null		now()
last_error	text			
ghl_id	text			

Column	Type	Null	Key	Default / references
created_at	timestamptz	not null		now()
sent_at	timestamptz			

ghl.review_queue

12 columns · no row-level security

Column	Type	Null	Key	Default / references
id	bigint	not null	PK	auto-increment
source	text	not null		
event_type	text	not null		
ghl_object	text			
ghl_id	text			
local_id	uuid			
summary	text	not null		
proposed	jsonb	not null		
state	review_state	not null		'open'
raised_at	timestamptz	not null		now()
decided_at	timestamptz			
decided_by	uuid			

ghl.reviewable_field

1 columns · no row-level security · Allowlist. An accepted CRM edit can write these and nothing else. Deliberately excludes every band 2 and band 3 column: a status change must not be a route to rewriting an address or a cost basis.

Column	Type	Null	Key	Default / references
column_name	text	not null	PK	

ghl.sync_state

7 columns · no row-level security

Column	Type	Null	Key	Default / references
resource	text	not null	PK	
location_id	text	not null		
cursor_at	timestamptz			
last_run_at	timestamptz			

Column	Type	Null	Key	Default / references
last_ok_at	timestamptz			
last_error	text			
records_seen	bigint	not null		0

ghl.transaction

17 columns · no row-level security

Column	Type	Null	Key	Default / references
ghl_id	text	not null	PK	
location_id	text	not null		
contact_id	text			
invoice_id	text			
subscription_id	text			
amount	numeric(12,2)	not null	CK	
currency	char(3)	not null		
amount_refunded	numeric(12,2)	not null	CK	0
status	text	not null		
live_mode	boolean	not null		
payment_provider	text			
entity_type	text			
entity_id	text			
ghl_created_at	timestamptz	not null		
ghl_updated_at	timestamptz	not null		
synced_at	timestamptz	not null		now()
raw	jsonb	not null		

ghl.webhook_event

9 columns · no row-level security

Column	Type	Null	Key	Default / references
webhook_id	text	not null	PK	
event_type	text	not null		
occurred_at	timestamptz	not null		

Column	Type	Null	Key	Default / references
received_at	timestamptz	not null		now()
processed_at	timestamptz			
attempts	integer	not null		0
last_error	text			
signature_ok	boolean	not null		
payload	jsonb	not null		

Schema gov

gov.data_right

15 columns · no row-level security

Column	Type	Null	Key	Default / references
right_id	text	not null	PK	
name	text	not null		
grantor	text	not null		
instrument	text	not null	CK	
source_code	text		FK	→ feed.listing_source
reference	text			
counterparty_contact	text			
effective_from	date			
effective_to	date			
survives_termination	boolean	not null		false
review_status	text	not null	CK	'unreviewed'
reviewed_by	text			
reviewed_on	date			
notes	text			
created_at	timestamptz	not null		now()

gov.data_right_territory

2 columns · no row-level security

Column	Type	Null	Key	Default / references
right_id	text	not null	FK PK	→ gov.data_right
territory_id	text	not null	FK PK	→ gov.territory

gov.data_right_use

4 columns · no row-level security

Column	Type	Null	Key	Default / references
right_id	text	not null	FK PK	→ gov.data_right
use_code	text	not null	FK PK	→ gov.use_kind
posture	text	not null	CK	'unclear'
condition	text			

gov.obligation

7 columns · no row-level security

Column	Type	Null	Key	Default / references
obligation_id	bigint	not null	PK	auto-increment
right_id	text	not null	FK	→ gov.data_right
kind	text	not null	CK	
interval_hours	integer			
text_required	text			
detail	text	not null		
enforcement	text	not null	CK	'unenforced'

gov.policy

5 columns · no row-level security

Column	Type	Null	Key	Default / references
id	boolean	not null	CK PK	true
enforcement_mode	text	not null	CK	'advisory'
changed_at	timestamptz	not null		now()
changed_by	text			
note	text			

gov.prohibited_dimension

4 columns · no row-level security

Column	Type	Null	Key	Default / references
dimension	text	not null	PK	
basis	text	not null		
kind	text	not null	CK	
rationale	text	not null		

gov.property_provenance

4 columns · no row-level security

Column	Type	Null	Key	Default / references
property_id	uuid	not null	FK PK	→ core.property
right_id	text	not null	FK PK	→ gov.data_right
scope	text	not null	CK PK	'listing_facts'
acquired_at	timestamptz	not null		now()

gov.regulation

9 columns · no row-level security

Column	Type	Null	Key	Default / references
reg_code	text	not null	PK	
name	text	not null		
citation	text			
regime	text	not null	CK	
applies_when	text	not null		
constrains	text	not null		
our_posture	text	not null		
status	text	not null	CK	'identified'
reviewed_on	date			

gov.regulation_control

4 columns · no row-level security

Column	Type	Null	Key	Default / references
reg_code	text	not null	FK PK	→ gov.regulation
control	text	not null	PK	
located_in	text	not null		

Column	Type	Null	Key	Default / references
kind	text	not null	CK	

gov.territory

7 columns · no row-level security

Column	Type	Null	Key	Default / references
territory_id	text	not null	PK	
kind	text	not null	CK	
name	text	not null		
country	char(2)	not null		'US'
state	char(2)		CK	
parent_id	text		FK	→ gov.territory
notes	text			

gov.territory_place

3 columns · no row-level security

Column	Type	Null	Key	Default / references
territory_id	text	not null	FK PK	→ gov.territory
city	text	not null	PK	
state	char(2)	not null	PK	

gov.use_kind

3 columns · no row-level security

Column	Type	Null	Key	Default / references
use_code	text	not null	PK	
label	text	not null		
detail	text	not null		

Schema intake

intake.batch

9 columns · no row-level security

Column	Type	Null	Key	Default / references
batch_id	uuid	not null	PK	gen_random_uuid()

Column	Type	Null	Key	Default / references
source_file	text	not null		
source_kind	text	not null	CK	'sdi_workbook'
mapping_version	text	not null		'v1'
right_id	text		FK	→ gov.data_right
uploaded_by	uuid		FK	→ core.person
uploaded_at	timestamptz	not null		now()
note	text			
status	text	not null	CK	'open'

intake.row

37 columns · no row-level security

Column	Type	Null	Key	Default / references
row_id	uuid	not null	PK	gen_random_uuid()
batch_id	uuid	not null	FK UQ	→ intake.batch
row_number	integer	not null	UQ	
raw	jsonb	not null		
street_address	text			
unit	text			
city	text			
state	char(2)			
zip	text			
property_type	text			
beds	smallint			
baths	numeric(3,1)			
sqft	integer			
year_built	smallint			
lat	numeric(9,6)			
lng	numeric(9,6)			
list_price	numeric(12,2)			
gross_rent_annual	numeric(12,2)			

Column	Type	Null	Key	Default / references
opex_annual	numeric(12,2)			
hoa_annual	numeric(12,2)			
market_rent_monthly	numeric(10,2)			
property_tax_annual	numeric(10,2)			
insurance_annual	numeric(10,2)			
maintenance_annual	numeric(10,2)			
management_fee_bps	integer			
vacancy_allowance_bps	integer			
lot_sqft	integer			
garage_spaces	smallint			
description	text			
internal_notes	text			
status	text	not null	CK	'pending'
problems	jsonb	not null		'[]'
reviewed_by	uuid		FK	→ core.person
reviewed_at	timestamptz			
review_note	text			
property_id	uuid		FK	→ core.property
released_at	timestamptz			

intake.zip_centroid

6 columns · no row-level security · Approximate ZIP centroids. Replace with a real geocoder before the address gate opens on anything that matters: once unlocked, the pin is only as accurate as this table.

Column	Type	Null	Key	Default / references
zip	text	not null	PK	
city	text	not null		
state	char(2)	not null		
lat	numeric(9,6)	not null		
lng	numeric(9,6)	not null		
source	text	not null		'approximate'

Appendix B. Change Log

Read from `CHANGELOG.md` at generation time, so this document cannot describe a release history the repository does not have. The current release level is `v0.9.13`, from `VERSION`.

> **Versions are 0.x deliberately: nothing here has run against a production GoHighLevel account or a live MLS feed, so the interfaces are not stable enough to promise compatibility. The first release that has is 1.0.0.**

0.9.13

2026-09-04

Favourites show the same photograph the search grid does.

Fixed

- The favourites list showed generated drawings while every other page showed photographs. The card image was chosen by a correlated subquery written inside the web tier's *search* query, so `api.my_favorite` — a different view — had no such column and the page fell back to the generated illustration. Nothing failed; one page just quietly showed something else.
- A test asserts the two agree, listing by listing. Written against `/api/listings` rather than `/api/view` — the latter is the older demo endpoint and returns neither `property_id` nor the card image, so a test against it would have proved nothing about either page.

Note

- `primary_image` deliberately lives on `api.property_card` and **not** on `api.property`, which would have been simpler. `core.property_media`'s row policy is written in terms of `api.property`, so a subquery over `api.property_media` inside `api.property` makes the policy depend on the view that depends on the policy — a recursion PostgreSQL discovers at query time, not at definition time.

0.9.12

2026-09-04

A property you can actually look at.

Added

- **Full screen.** A ■ button beside the close button widens the detail panel to the whole window. Not simply a wider column: above 1000px the photographs take the left and the figures the right, so the space goes to the pictures rather than to a 1600px line of text. The choice is remembered per browser — somebody who wants the big view wants it for every listing.
- **See all N photos**, on the lead image. A full-window page of every photograph, captioned, four across, with a *shows the street* badge on anything gated.
- **A large single-photograph view.** Click any tile — or any image in the panel — for the full file with a caption and an *n of N* counter. Arrow keys move and wrap in both directions.
- Escape unwinds **one layer at a time**: large view → grid → listing. Closing everything at once loses both the photograph and the listing it came from.

Notable

- Photographs are grouped by caption, and the **headings appear only when they group more than one image**. With one photograph per room — every listing today — a heading above each is a full-width divider that forces one tile per row and says nothing the caption does not. They return the moment a listing has several kitchen shots. Grouping is by caption rather than filename because the caption comes from the database and is the field staff will

edit; filename sniffing would work for the seeded assets and break for anything served from the media store, where the name is a uuid.

Fixed

- `.lightbox{display:flex}` overrode the `hidden` attribute — an author rule setting `display` beats the browser's `[hidden]{display:none}` — leaving a transparent full-window layer over the page that swallowed every click. The listing cards simply stopped responding and nothing looked wrong. Caught by driving the page in a real browser, not by reading it.
- Section 8 of the Feature Test Plan was stale: it still said three photographs per listing and told the reader to look for the word ILLUSTRATION, which has not been true since the supplied photography landed.

0.9.11

2026-09-04

The supplied interiors actually reach the listings.

Added

- `33_interior_media.sql` — each listing's living room, kitchen and bedroom, paired by the same deterministic `listing_ref` ordering the exteriors use, so a listing's card and its interiors stay together across a rebuild.
- 720px thumbnails for all 75 (15.4 MB → 3.6 MB), used in the gallery strip.

Changed

- **The gallery showed only three images.** `media.slice(1, 3)` was fine when everything after the card was a drawing and stopped being fine the moment real interiors arrived — the bedroom never appeared and nothing said so. It now renders every photograph, in a strip that wraps to however many there are, instead of a two-column grid built around exactly three.
- The generated `hero.svg` at position 0 is removed. It was the card image until `30_stock_media.sql` put a photograph in front of it; since then it has been a drawing sitting between a real photograph and three real interiors. The renderer still draws one on demand, so the client-side fallback for a file that will not load is unaffected.
- Captions read *"Living area — representative photo, not the actual property"*. Same reason as the exteriors: an investor who walks the house and finds a different kitchen stops trusting the numbers too.

Notable

- These stay **static assets in the image**, not rows in the media store built in 0.9.8. The store is for operational photography that staff add and manage; seeding a demo into it would make its contents a mixture of things that can be purged and things that come back on the next deploy.
- `reveals_location` is false on all of them. The flag means *identifying*, not *interior* — a photograph of a different kitchen identifies nothing.

0.9.10

2026-09-04

Say which CPU feature is missing, instead of a stack trace about `endsWith`.

Fixed

- `scan-media.js` reported `Cannot read properties of undefined (reading 'endsWith') when sharp` would not load. That message comes from **sharp's own error handler**: it collects the reasons the load failed and then formats them with `err.code.endsWith(...)`, and one of the collected errors has no `code`, so the formatter throws and takes the real reason with it.

Worth noting

- The build-time assertion added in 0.9.9 did its job and still could not catch this. It runs on a GitHub runner with a modern CPU, so sharp loads there. No image-side check can see a missing CPU feature on a machine it will never run on — which is the argument for the runtime message naming it precisely.

0.9.9

2026-09-04

A worker image that can actually load its image library.

Fixed

- The published worker image installed `sharp` cleanly and then could not load it, so `scan-media.js` refused to run. `sharp` ships prebuilt native binaries as per-platform optional dependencies, and **musl is a different binary from glibc**. The image was built `FROM node:22-alpine` (musl) while the lockfile and every test were made on glibc, so `npm ci` resolved a binary the runtime could not use. The worker now builds `FROM node:22-slim`, matching the platform the lockfile and the tests were made on.
- The Dockerfile now **asserts sharp loads and encodes a JPEG at build time**, so an image that cannot do its job fails the build instead of being published. This is the same failure mode as the earlier web image that copied two of its four modules: passes every test, breaks only in the registry.
- `scan-media.js` printed *"sharp is not installed"* for every failure, including a native binary that was present but would not load. It now prints the real error and says plainly that this is a broken image rather than a configuration problem — the wrong message sent the diagnosis in the wrong direction.

0.9.8

2026-09-04

Photographs stop being part of the build.

Added

- **A media store on a shared mount**, outside the image. A host path rather than a Docker named volume, because `down -v` destroys named volumes and the schema flow needs `down -v`; photographs must not die with a rebuild. Four zones: `inbox/` (staff drop files here), `store/` (machine-managed), `quarantine/`, `purged/`.
- `/media/file/<media_id>` — the authorising route. It re-asks the database, *as the caller*, before any bytes leave. A row the caller cannot see is a **404, not a 403**: a 403 would confirm the photograph exists, which is half of what the gate withholds.
- `scan-media.js` — ingest, reconcile and purge. Ingest re-encodes every file, which is what strips EXIF, then thumbnails, stores under an id and registers it as pending.
- `core.media_event` — who published what, and when.
- **Lifecycle columns** on `core.property_media`: `storage_path`, `state`, `published_at`, `purge_after`, `legal_hold`, `content_sha256`.
- 17 tests for the above, including one that builds a JPEG carrying GPS coordinates and asserts they are gone from the stored bytes.

Notable decisions

- **The filename in `store/` is the `media_id`.** Anyone holding a file identifies it with one query and no lookup table; `ls store/SDI-1009/` is the complete set for a listing. The row still stores its own `storage_path`, because a listing reference can be corrected and a derived path silently stops resolving when it is.
- **EXIF stripping is not housekeeping.** A phone photograph carries GPS accurate to a few metres. Untouched, the exact address sits inside the file — the gate intact in the database and bypassed in fact. It runs on the share path too, because a file copied from a PC never passed through a browser.

- **Arrival fails closed.** Every photograph lands pending, unpublished, and `reveals_location = true`. Publish-and-correct-later leaves the gate open between arrival and review, and no correction un-discloses anything.
- **Deletion happens twice.** Unpublish is immediate and reversible; destruction waits out a retention window. Legal hold beats a due retention date, or a well-behaved cleanup job destroys exactly the evidence somebody needs.
- **Reconcile reports and fixes nothing.** A row with no file may be a restore that failed; a file with no row may be the only copy of something.
- **The scanner can register but not publish.** No `EXECUTE` on `api.media_publish` for `sdi_integration` — an unattended process that can publish defeats the review step entirely.

Fixed during the build

- The route first read `core.property_media` directly. The reader roles hold `SELECT` on that table but **no** `USAGE` **on schema** `core`, so it would have 404'd every photograph for everyone except an admin. Now read through `api.media_bytes`.
- The scanner first read `core.property` directly. Every policy on that table is scoped `TO` a named application role and `sdi_integration` is not one, so the read returned **zero rows rather than an error** — every folder reported as an unknown listing while the scanner looked like it was working. Now `api.listing_id()`.
- Authorisation for the scanner was first a `session_user = 'sdi_integration'` special case, which could not be tested without connecting as that exact role. Replaced with a service account person, so the scanner authorises through the same predicate as everyone else and the audit trail names it.

0.9.7

2026-09-04

The media lifecycle, written down before it is built.

Added

- `docs/Property-Media-Lifecycle.pdf` (13pp) — the operational requirements for listing photography: how a file arrives, how it acquires meaning, how it is maintained, and when it is destroyed. Section 9 states plainly which parts exist today, so a requirements document is not mistaken for a status report.

Fixed

- The document list in `README.md` was stale: it named two of the six documents, gave the wrong page count for one, pointed at a renamed runbook, and said 63 worker tests when there are 70.

0.9.6

2026-09-04

A hovered map pin you can read.

Fixed

- Hovering a listing on the map turned the pin's background dark but left the price in the dark slate that `.pin.gated` sets for a white background — 1.6:1 contrast, effectively invisible. `.pin.hi` now restates the foreground as white (9.4:1). It sets `border-color` rather than `border`, so a gated pin keeps its dashed edge; that dash is what says the position is approximate. The SVG fallback map already did this correctly, which is why it only showed on the Leaflet map.
- The web test suite skipped every database test when PostgreSQL was unreachable, and reported `34 tests, 0 fail` with twenty-seven of them never run. That reads as green. The skip is now opt-in — `SDI_TEST_NO_DB=1` for the pure-logic tests on a machine with no database, and a hard failure naming the fix otherwise.

0.9.5

2026-09-04

The photographs, at a weight a page can carry.

Added

- `core.property_media.thumb_url`: a downscaled copy of the same picture, 720px and ~84 KB. Null means there is no smaller version, so every reader coalesces rather than assumes. It carries no visibility of its own — it is the same photograph, released or withheld with the row that owns it, so a thumbnail is never a way around a gated image.
- `web/public/assets/thumb/` — the 25 downscaled copies. 9.4 MB → 2.1 MB.
- An image fallback in `app.js`: a media row can outlive the file it names (108 Fairgrove's `front` is seeded before the photograph has been supplied, and a mistyped path looks the same), so a failed load falls back to the generated illustration instead of a broken-image icon.

Changed

- `/api/listings` returns `coalesce(thumb_url, url)` as `primary_image`; the detail gallery keeps the full file for the lead image and uses thumbnails for the strip beneath it.

Fixed

- The 25 photographs are now committed. They had been placed on the deployment host only, and `web/Dockerfile` copies `public/` at build time — so the published image would have carried none of them.

0.9.4

2026-09-04

Supplied photography on the cards.

Added

- `30_stock_media.sql` points each of the 25 listings at one of the supplied photographs, paired deterministically by listing reference so a rebuild does not reshuffle them.
- `/api/listings` now returns `primary_image`, read through `api.property_media` rather than naming a file — so a gated photograph can never become a card thumbnail by accident. The generated illustration remains the fallback for any listing without one.
- `gov.data_right` `STOCK-PHOTOGRAPHY`, recorded **unreviewed**: the source and licence of the supplied images are not established, and a stock licence, an attribution-required licence and a broker's listing photograph look identical on disk.

Changed

- These images are **not photographs of these properties**, and that one fact decides two things. `reveals_location` is false, because the flag means *identifying*, not *exterior* — a photograph of a different house identifies nothing. And every one is captioned as representative, because a stock exterior presented without qualification reads as a picture of the property.

0.9.3

2026-09-04

A cooler palette, and cards that look like different properties.

Changed

- Palette moved from a neutral ground with amber accents to a light blue one. At card size the warm accents read as brown, and twenty of them in a grid looked tired. The gated state is now a soft steel blue rather than rust — kept clearly lighter and greyer than the interactive accent, because if "locked" and "clickable" look alike the banner stops meaning anything.
- Listing illustrations redrawn in daylight colours: cool siding, slate roofs, pale interiors with one saturated accent each.

Added

- A card hero keyed to **property type** — a duplex has two doors, a condo is a stack — and varied by property id, so a grid of twenty looks like twenty properties rather than one repeated image. Deliberately generic: no street, no number, no surroundings, so it is not location-revealing and is not gated. The real exterior stays behind `reveals_location`.

Fixed

- `docs/gen_detail.py` crashed on listings with null sizes — the tracked Irvine address and any unfilled workbook import. It now skips them rather than inventing figures, which is what the rest of the system exists to prevent.

0.9.2

2026-09-03

A duplicate-listing bug, found by loading the same workbook twice.

Fixed

- Loading a workbook twice and releasing both batches created **four listings for two addresses**. Validation ran at load time only, so a row that was clean when the file arrived stayed "clean" through approval and release even after the same address had been released from another batch.
- The first attempt at that fix silently did nothing. `SELECT * INTO r FROM intake.row WHERE row_id = r.row_id` self-references the record being overwritten: `r` is cleared as the assignment begins, the `WHERE` matches nothing, `r` comes back NULL, and every test on it is NULL rather than false — so the guard never fired and the insert proceeded into a constraint violation instead of a clean skip. The row id now goes into its own variable first.

Added

- `docs/Design-Conflict-Register.pdf` — the KAVADOO design document (v1.0, April 2026) compared against the build. **12 conflicts** where the two specify different things, kept separate from **16 gaps** (specified, not built) and **8 unspecified** items (built, not mentioned). A gap needs scheduling; a conflict needs a decision, and mixing them buries the decisions.
- `ux_property_live_address` — a partial unique index on `core.property`, so no code path can list the same address twice while it is live. Scoped to `draft`, `active`, `coming_soon` and `pending`, because an address genuinely recurs across years as it sells and is relisted. Re-validating closes the path the intake code owns; the index closes the others.
- A regression walkthrough in `29_intake_tests.sql` covering exactly the sequence that produced the duplicates.

0.9.1

2026-09-03

Fixes found deploying 0.9.0.

Fixed

- The intake loader failed with `permission denied for schema api`. The role it runs as, `sdi_integration`, was granted SELECT on the two views it reads but never USAGE on the schema holding them, which is a no-op that fails at run time. Same for `gov`, one error later.
- The start-up fair-housing check retried for only ~25 seconds. A first boot that loads the whole schema takes longer, so the web container hit its FATAL path and recovered only because of the restart policy. `depends_on: service_healthy` does not help: the postgres entrypoint runs init against a unix socket, so the healthcheck passes while TCP is still refused. Budget is now ~2 minutes.

Added

- `docs/Deployment-Runbook.pdf` — the numbered procedure, with the expected output at each step and a troubleshooting table built from what actually went wrong on the host.

0.9.0

2026-09-02

Spreadsheet intake, the review queue, and a test plan.

Added

- Schema `intake`: staged property loads. `intake.batch` (one file, one upload), `intake.row` (one property, holding both the verbatim payload and our reading of it), `intake.zip_centroid`.
- `tools/workbook-to-json.py` — reads the SDI analysis workbook's `Import` sheet by label and emits intake JSON. Python, because it needs a spreadsheet library the worker deliberately does not carry.
- `worker/tools/load-intake.js` — loads that JSON, validates every row, and prints what a reviewer has to decide on.
- `/admin.html` — the staff review screen. Approve, reject and release, by row or by batch, with the verbatim payload one click away.
- `api.review_intake_rows()`, `api.approve_batch()`, `api.release_intake_rows()`, `api.release_batch()`; views `api.intake_batch` and `api.intake_row`.
- `docs/Feature-Test-Plan.pdf` — 23 pages of acceptance tests, including a section whose every test is expected to be refused.
- `docs/Deployment-Runbook.pdf` and `docs/DEPLOY-RUNBOOK.md` — the numbered deployment procedure, with the expected output at each step, verified against the live host rather than written from the compose file. Includes a troubleshooting table mapping each symptom seen during deployment to its actual cause.
- `VERSION`, `CHANGELOG.md`, and System Documentation section 15, *Build and Release*. Both PDFs read the version and change log at generation time, so they cannot describe a release history the repository does not have.
- System documentation section 10, *Getting Properties In*.

Changed

- Validation splits blocking errors from warnings. A reviewer forced to clear every oddity before releasing anything stops reading the oddities.
- Release records provenance against `gov.data_right` `SDI-WORKBOOK`, recorded **unreviewed**: the financial modelling is SDI's own, but the property description in the same file is verbatim MLS listing copy whose republication right is not established.

Fixed

- `api.release_intake_rows()` did not close a batch it had emptied, so a batch released row by row read "open" forever.
- The review screen's select-all box kept its checked state across the redraw that follows every action while the selection was cleared, so the next click unselected everything.

- Worker integration tests proved the fee gate *opens* and never shut it again, leaving Marcus's gate open and silently destroying the demo's central contrast on every test run. Now restores the fixture and asserts the restore worked.

Security

- The workbook's "Schools Rating" and its composite deal score are kept in the raw payload and never promoted to a column. Both are registered fair-housing proxies; `api.security_invariants()` fails if either name appears in `core` or `api`.

0.8.0

2026-09-02

Data rights, territory, and regulatory compliance.

Added

- Schema `gov`: `data_right`, `data_right_territory`, `data_right_use`, `obligation`, `property_provenance`, `territory`, `regulation`, `regulation_control`, `prohibited_dimension`, `policy`.
- `gov.may_use(property_id, use, scope)` — the single question the publication path asks. Confirmed, unexpired, in territory, and granting this use: all four, not a score.
- Register of 17 regulatory regimes with the trigger condition for each and where the constraint is actually enforced. Ten currently have no control and say so.
- `docs/data-rights-intake.md` and `worker/tools/record-data-right.js`.
- System documentation section 9, *Data Rights and Compliance*.

Changed

- Governance ships in **advisory** mode. A control that refused to publish without a confirmed right would take a working marketplace off the air over paperwork nobody has transcribed yet. Flipping `gov.policy` to `blocking` is the go-live gate; the standing invariant then keeps it shut.
- `api.security_invariants()` gained three governance checks.

Security

- Fair housing enforced three ways: the prohibited-dimension register is a table, the standing invariant fails if any listed dimension is exposed as a readable column, and the web tier refuses to start if its filter allowlist intersects the register.

0.7.0

2026-09-01

Listing status tracked against an external source.

Added

- Schema `feed`: `listing_source`, `property_external`, `observation`, `status_change`, `status_map`, `review_flag`, `manual_check`.
- `feed.observe()` and `feed.reconcile()`. An observation never writes a status directly; it is recorded, then reconciled.
- Four source adapters behind one interface: manual, RESO Web API, RentCast, and a consumer portal left deliberately unimplemented.
- `worker/tools/check-listings.js` — the nightly sweep, oldest check first.
- System documentation section 8, *Where Listing Data Comes From*.

Changed

- Two deliberate asymmetries: an `error` outcome is never read as an absence, and a listing returning to market is acted on the **first** sighting because escrow fails.

0.6.0

2026-09-01

The marketplace.

Added

- Map-based browsing: filters across the top, map on the left, cards on the right, drill-down panel over it.
- `core.market_area`, `core.property_detail`, `core.property_media`, `core.saved_search`; views `api.property_detail`, `api.property_media`, `api.my_favorite`, `api.my_saved_search`.
- Favourites, saved searches, and a plain-English search box (a rules parser, not a model — the seam and its validator are what had to be right first).
- `web/media.js` — deterministic generated listing illustrations, so no stock service and nobody's copyright.
- 16 further demo listings spanning 1–6 beds, $87k–$581k, nine cities.
- System documentation section 7, *The Marketplace*.

Security

- Photographs inherit the address gate. `property_media.reveals_location` marks exterior views, released on the same predicate as the street address — without it the gate would reopen through the picture gallery.
- Gated listings are drawn on the map as a fuzzed ring rather than a pin, because their coordinates are deliberately offset.

0.5.0

2026-09-01

Authentication.

Added

- `core.credential` and `core.session`, both RLS-forced with no policy at all, reachable only through `SECURITY DEFINER` functions.
- `scrypt` password hashing, session cookies, a login page.
- Account lockout after five failures; a password change revokes every session.

Changed

- Nothing beneath it. Not one policy, view or grant was modified to add authentication — the database contract was always "here is a role and an actor id", and a session now supplies those instead of a dropdown.
- The demo persona switcher is off unless `DEMO_PERSONAS=1`.

0.4.0

2026-08-31

Deployable from published images.

Added

- `.github/workflows/release.yml` — publishes `db`, `web` and `worker` to GHCR.
- `docker-compose.release.yml` for pulling by tag; Portainer instructions.

Changed

- Role credentials are taken from the environment at first start rather than baked in. A role given no password stays `NOLOGIN`, failing at connect time rather than quietly granting access.

0.3.0

2026-08-31

Review queue, deal pipeline, and the system documentation.

Added

- `core.review_queue` and review actions with an allowlist on accepted CRM edits.
- Deal pipeline, stage history and per-party visibility.
- `docs/System-Documentation.pdf`, generated from the repository and a freshly built database rather than written from memory.

0.2.0

2026-08-31

The GoHighLevel bridge.

Added

- Schema `ghl`: id mapping, webhook events, transactions, fee agreements, sync state, outbox.
- Integration worker: webhook intake with signature verification, outbox drain, document and transaction sync, EspoCRM migration passes.
- `docs/GHL-Interface-Specification.pdf`.

Security

- The fee gate requires **both** a completed document and a paid transaction. A signed-but-unpaid document does not open it.

0.1.0

2026-08-31

Foundation: the visibility model.

Added

- PostgreSQL schema with three visibility bands enforced by row-level security and column grants, not by application code.
- Four-schema separation: `core` (tables, no app-role access), `sec` (predicates), `api` (masking views), plus the roles that use them.
- `api.security_invariants()` — must always return zero rows.
- Demo dataset and the Marcus/Ruth pair: same role, same query, one timestamp of difference.